Stat 414 Notes

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Course Goals for STAT 414

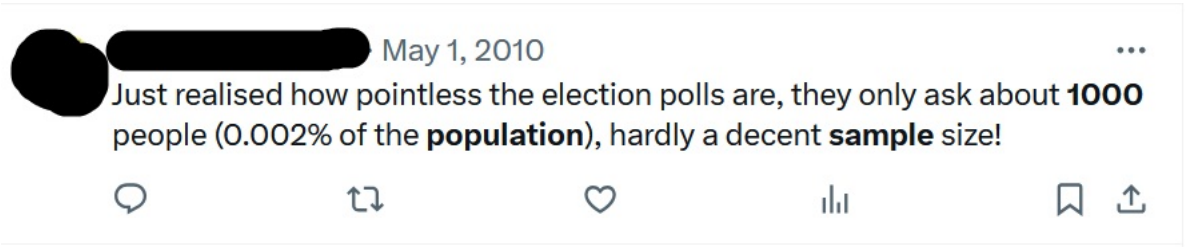
STAT 414 is an introduction to sampling. By the end of the course, you will:

- Learn the basic methodology of how classical survey samples are conducted based on simple random sampling.
- Recognize sampling and non-sampling errors.
- Be able to use extra information in sampling via ratio estimators and regression estimators.
- Be able to define and use stratified samples and know why they are often better than SRS.
- Be able to design one and two stage cluster samples and know their key properties.
- Recognize strengths and weaknesses of different sampling approaches.
- Use standard R packages to analyze survey data and estimate population means, proportions, totals, and more.

1 Introduction to Sampling

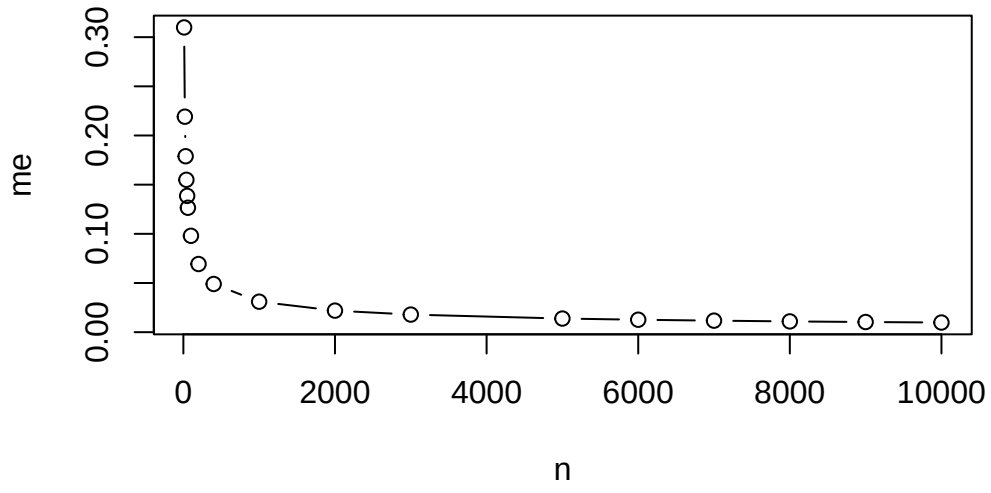
1.1 Introduction

If you spend any time at all on social media (especially in an election year), you're likely to come across a post like this:



What do you think? Do you agree?

```
n<-c(10,20,30,40,50,60,100,200,400,1000,2000,3000,5000,6000,7000,8000,9000,10000)
me<-1.96*sqrt(0.25/n)
plot(n,me,type="b")
```

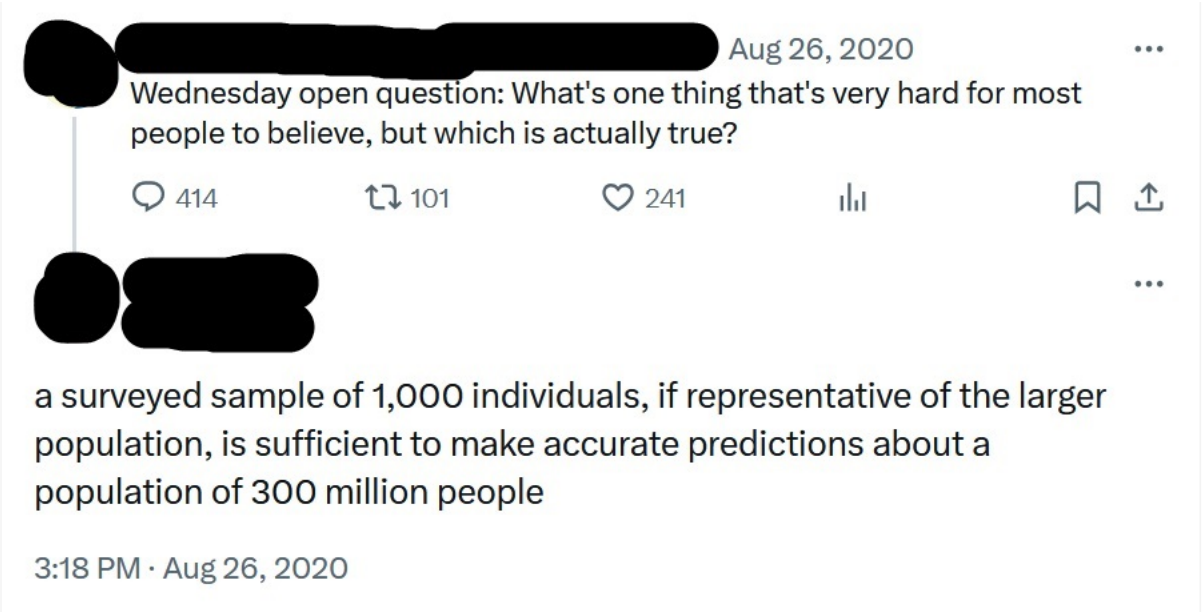


So why even sample?

1. Cheaper

2. Faster

3. **More Accurate** (?!?!?)



What's key here?

What we will spend this semester studying is how to ensure this, to make sure we can make accurate predictions. This will involve answering questions focused on:

- Survey design
- Preparation
- Collect and prepare data
- Data analysis

How is sampling different from previous courses, like STAT 212?

How is sampling different from previous courses, like STAT 262 or STAT 380?

How is sampling different from previous courses, like STAT 102 or STAT 318?

However, like previous courses, sample surveys do have an overarching process, some of which overlap with steps/processes we've seen before. We'll go through the principal considerations in a sample survey, along with some definitions and (possibly) new notation.

1.2 Considerations in planning and executing a survey

1. **Objectives of the survey**
2. **Defining the population** In sampling, the word **population** needs some clarification. Generally, it means the larger set from which the sample is chosen, and it's the larger group we wish to make inference/draw conclusions about. And a **sample** is a subset of the population. However, we must distinguish between

- **Target population**

- **Sampled population**

Ideally, these two overlap completely, but this does not always occur. Plus, we also need to consider the **sampling frame population**

I love the figure on page 5 of the book:

Within the **sampling frame** we have

- **Observation unit**

- **Sampling unit**

For example,

3. **Identifying potential sources of bias and ameliorating them**: In previous classes, like STAT 262 or STAT 380, we use the word **bias** to mean an estimator that systematically over- or under-estimates the true population characteristic. In sample surveys, we still use the same definition, but now biased estimators typically arise for two reasons:

- Population characteristic of interest (Estimation bias):

- Selection and/or measurement bias: Selection bias occurs when there is a mismatch between the sampled population and the target population. Measurement bias occurs when the survey observation tends to differ from the truth in a particular direction.

We'll talk about each of these in more detail.

Selection bias

- Convenience samples
- Judgment samples
- Self-selected samples
- Undercoverage
- Overcoverage
- Nonresponse

Measurement bias

Why can it occur?

- Unavoidable Reasons

- People!

- Survey mode: method used to distribute and gather responses

So we're concerned about response bias–what now? Careful question design is key. We could spend weeks on just questionnaire design, but will only do a very brief overview. There are other resources on campus that can help with this aspect of surveys (Bureau of Sociological Research).

We'll start with types of questions to avoid, and then move on to other considerations when writing questions.

Common types of biased questions:

- Leading questions

  How to fix it:

- Assumptive questions

  How to fix it:

- Double-barreled questions

  How to fix it:

- Double negatives

  How to fix it:

Other considerations:

- Pay careful attention to answer scale options

- Check formatting

- Avoid jargon, and be specific

- Open or closed questions?

- Question order matters

Probably most important, test questions ahead of time with the target population.

4. **Choosing the sampling design/data collection approach**

5. **Choosing the analysis approach**

6. **Data analysis**

7. **What's next?** The more information we have about a population, the easier it is to devise a sampling plan that will give accurate estimates. Any past survey can serve as a guide for future samples, as it will provide information about variablity and cost.

The rest of the semester

2 An Introduction to Probability Samples

Before we can jump into sampling designs, we need to establish a few basic concepts and notation. The sampling designs we're going to discuss are **probability samples**.

Definition: Probability sample

A **probability sample**

Throughout this semester we'll discuss three different ways to design a probability sample, as well as the properties of estimators arising from these designs. Probability samples allow us to avoid selection biases and help ensure the sample is representative of the population, as probability samples take selection choice away from the researcher, and put it in the hands of random selection. The three major designs we'll discuss are:

- Simple random sampling
- Stratified random sampling
- Cluster sampling

We'll also consider **systematic sampling**

To simplify things, we'll make the following assumptions:

Depending on how far we get, we may revisit relaxing some of these assumptions at the end of the semester.

In all cases, our goal is to learn about some characteristic of the target population

and we'll use our sample data to estimate that characteristic. We'll also estimate standard errors, and use those to construct confidence intervals for the unknown population characteristic. We'll also discuss determining the necessary sample size.

2.1 Probability Sampling Definitions and Notation

Throughout this semester, we'll assume a finite population, or **universe** of N\$N\$\$N\$\$N\$\$N\$ units:

Any sample we select from $\mathcal{U}$\$\mathcal{U}\$ will be a subset of $\mathcal{U}$, and we'll denote the specific sample chosen as $\mathcal{S}$. $\mathcal{S}$ contains n\$n\$ elements from $\mathcal{U}$.

Example: Suppose $\mathcal{U} = \{1, 2, 3, 4, 5\}$ and we consider samples of size $n = 3$. There are 10 possible samples that could be selected:

Remember that in **probability sampling**

Because each possible sample $\mathcal{S}$ has a known probability of being selected, each unit in the universe $\mathcal{U}$ also has a known probability of being in the selected sample.

Let y_i be

We consider y_i to a be a *fixed quantity.*

2.2 Probability Samples: Inference Basics

We want to use the data from a sample to estimate some characteristic of the population. Common population characteristics of interest are

Example: Earlier we considered the universe $\mathcal{U} = \{1, 2, 3, 4, 5\}$. Suppose the values of y_i are

i	1	2	3	4	5
y_i	2	4	4	6	6

We also considered taking samples of size $n = 3$ from $\mathcal{U}$. There are 10 possible samples, and all are equally likely.

Just like in other classes like STAT 102, 380, 212 our inference methods are going to rely on the **sampling distribution** of a statistic.

Definition: Sampling Distribution

The **sampling distribution**

Example, cont.:

From the sampling distribution, we can find properties of the estimator we're using.

- **Expected value**:

- **Estimation bias**:

- **Variance**:

- **Mean squared error**:

3 Simple Random Samples

Simple random sampling (SRS) is the most basic probability sampling design, and it is also used in combination with other designs like stratified and cluster sampling.

Definition: Simple Random Sample

A **simple random sample (SRS)**

This definition implies

Specifically, the **inclusion probability** for any sampling unit is

Note that the converse is not true!

This is called an **equal probability** design, and the SRS is only one example of equal probability design. Some stratified and cluster sampling designs also have an equal inclusion probability for all sampling units.

3.1 SRS: Selecting the Sample

There are two types of selection procedures, depending on whether or not the sampling units are returned to the population after they have been drawn:

- **SRSWR**

- **SRSWOR**

Conceptually, an SRS (SRSWOR) is selected as follows:

-

-

-

-

  ⋮

-

This is useful to see why each unit has the same inclusion probability.

But, this is not an especially realistic way to pick a sample. In practice, an SRS (SRSWOR) is selected as follows:

- Create a sampling frame

- Determine a selection procedure that performs SRSWOR

- A sampling unit is included in the sample if its index corresponds to one of the n random numbers selected.

So what are some selection procedures that perform SRSWOR?

- Lottery

- random.org

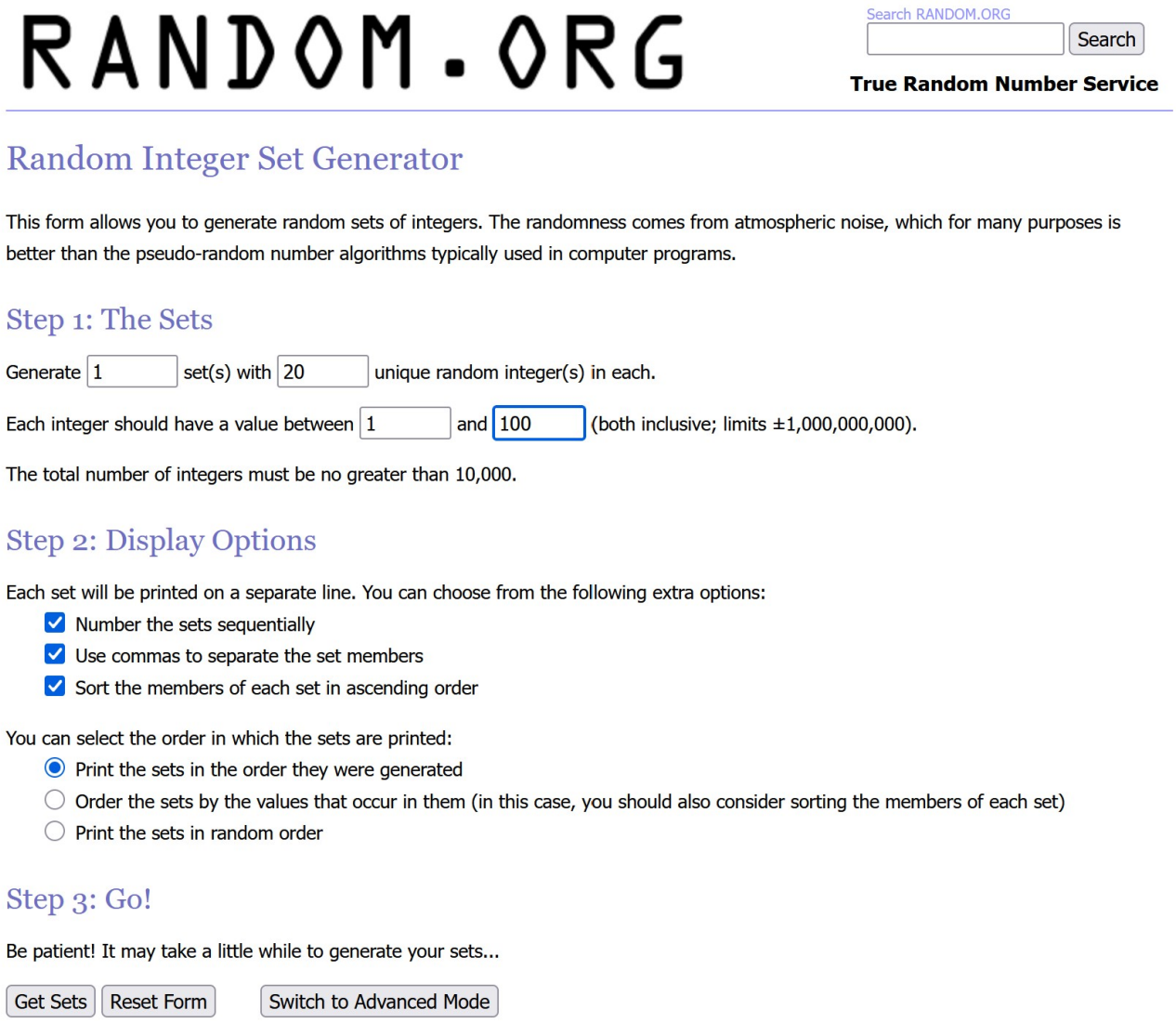


Figure 3.1: Random.org settings to generate a SRSWOR

- Statistical software
  - SAS: `proc surveyselect method=srs;`
  - Base R: `sample(1:N, n, replace=FALSE)`
  - R packages: Need to install and load the package `sampling`, then `sample<-srswor(n,N)`. You can then use the `getdata` function to extract the sample from the population.

Example: The dataset `baseball.csv` in Canvas contains statistics on 797 baseball players from the rosters of all major league teams in November, 2004. We'll treat this as our population, and take an SRS of 150 players.

First, we'll read the data into R, print out the first few rows, and check the size of the population:

```
baseball<-read.csv("baseball.csv",header=TRUE)
head(baseball)
```

```
  team leagueid   player   salary pos gplay gstart inning putout assist error
1  ANA       AL anderga0  6200000  CF   112     92   2375    211      5     2
2  ANA       AL colonba0 11000000   P     3     34    625      8     30     3
3  ANA       AL davanje0   375000  CF   108     27    743     75      1     0
4  ANA       AL donnebr0   375000   P     5      0    126      2      2     0
5  ANA       AL eckstda0  2150000  SS   142    136   3575    198    309     6
6  ANA       AL erstada0  7750000  1B   125    124   3196    986     66     4
  dplay pball gbat atbat run hit secbase thirdbase homerun rbi stolenb csteal
1     1    -9  112   442  57 133      20         1      14  75       2      1
2     4    -9    3     3   0   0       0         0       0   0       0      0
3     1    -9  108   285  41  79      11         4       7  34      18      3
4     0    -9    5     0   0   0       0         0       0   0       0      0
5    75    -9  142   566  92 156      24         1       2  35      16      5
6    83    -9  125   495  79 146      29         1       7  69      16      1
  walk strikeout intwalk hbp sacrhit sacrfly gidplay
1   29        75       6   1       0       3       3
```

```
2    0         1       0   0       0       0       0
3   46        54       2   0       1       5       2
4    0         0       0   0       0       0       0
5   42        49       1  13      14       2      11
6   37        74       1   4       3       4       9
```

```
N<-nrow(baseball)
N
```

```
[1] 797
```

Now we'll load the necessary package, set a seed and draw a sample of 150 players:

```
library(sampling)

set.seed(98263476)
n<-150
index<-srswor(n,N)
index[1:10]
```

```
 [1] 0 0 0 0 0 0 0 1 0 1
```

So the `srswor` function only gives us a 0/1 for each element in the population. To get the sample, there are a couple of options:

```
baseballsrs<-getdata(baseball,index)
head(baseballsrs)
```

```
   ID_unit team leagueid   player  salary pos gplay gstart inning putout assist
8        8  ANA       AL figgich0  320000  3B   148     80   2116     57    129
10      10  ANA       AL greggke0  301500   P     5      0    263      2      5
21      21  ANA       AL rodrifr0  375000   P     3      0    252      5      9
31      31  ARI       NL cintral0  335000  SS   154    125   3297    141    383
33      33  ARI       NL daiglca0  300000   P    11     10    147      5      6
50      50  ARI       NL sexsori0 8725000  1B    23     23    613    199     27
   error dplay pball gbat atbat run hit secbase thirdbase homerun rbi stolenb
8     11     9    -9  148   577  83 171      22        17       5  60      34
10     0     1    -9    5     0   0   0       0         0       0   0       0
21     0     0    -9    3     0   0   0       0         0       0   0       0
31    15    61    -9  154   564  56 148      31         7       4  49       3
33     0     0    -9   11    17   2   2       2         0       0   0       0
50     1    19    -9   23    90  20  21       4         0       9  23       0
```

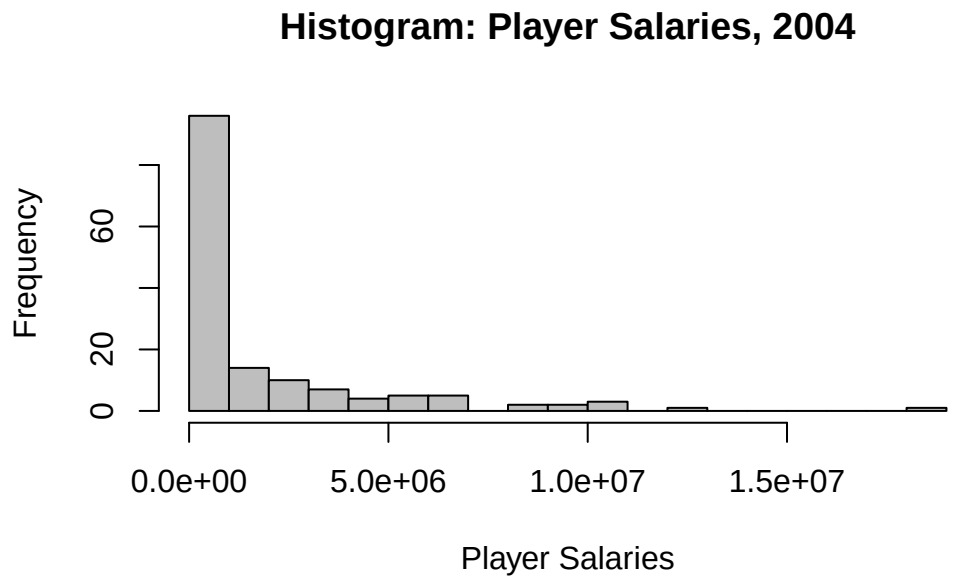
```
   csteal walk strikeout intwalk hbp sacrhit sacrfly gidplay
8      13   49        94       0   0      10       2       6
10      0    0         0       0   0       0       0       0
21      0    0         0       0   0       0       0       0
31      3   31        59       2   2      12       4      11
33      0    0         7       0   0       1       0       1
50      0   14        21       0   0       0       0       2
```

```
baseballsrs2<-baseball[(1:N)[index==1],]
head(baseballsrs2)
```

```
   team leagueid   player  salary pos gplay gstart inning putout assist error
8   ANA       AL figgich0  320000  3B   148     80   2116     57    129    11
10  ANA       AL greggke0  301500   P     5      0    263      2      5     0
21  ANA       AL rodrifr0  375000   P     3      0    252      5      9     0
31  ARI       NL cintral0  335000  SS   154    125   3297    141    383    15
33  ARI       NL daiglca0  300000   P    11     10    147      5      6     0
50  ARI       NL sexsori0 8725000  1B    23     23    613    199     27     1
   dplay pball gbat atbat run hit secbase thirdbase homerun rbi stolenb csteal
8      9    -9  148   577  83 171      22        17       5  60      34     13
10     1    -9    5     0   0   0       0         0       0   0       0      0
21     0    -9    3     0   0   0       0         0       0   0       0      0
31    61    -9  154   564  56 148      31         7       4  49       3      3
33     0    -9   11    17   2   2       2         0       0   0       0      0
50    19    -9   23    90  20  21       4         0       9  23       0      0
   walk strikeout intwalk hbp sacrhit sacrfly gidplay
8    49        94       0   0      10       2       6
10    0         0       0   0       0       0       0
21    0         0       0   0       0       0       0
31   31        59       2   2      12       4      11
33    0         7       0   0       1       0       1
50   14        21       0   0       0       0       2
```

Let's plot the salaries from our sampled players

```
hist(baseballsrs$salary,breaks=20,col="gray",xlab="Player Salaries",
   main="Histogram: Player Salaries, 2004")
```



3.2 Estimation from a SRS

Once we've selected our sample, we can use it to estimate characteristics of the population, along with their standard errors.

3.2.1 Estimating a population mean

As a reminder, the population mean is

A natural estimator of $\bar{y}_{\mathcal{U}}$ using a SRS of sample size n is the **sample mean**:

Example: In our sample of $n = 150$ baseball players, the mean salary is

```
mean.salary<-mean(baseballsrs$salary)
mean.salary
```

```
[1] 1993886
```

Because we have access to the entire population, we can check to see how our sample mean compares to the population mean.

```
mean.pop<-mean(baseball$salary)
mean.pop
```

```
[1] 2497669
```

What happens if we take a different sample? Let's try a few. What do we need to change in R?

It turns out that with simple random sampling, the sample mean is an unbiased estimator of the population mean. That is,

As noted in Chapter 2, the finite population variance is

Under simple random sampling, the sample variance s^2

is an unbiased estimator of σ^2. However, if we want to use $\bar{y}$ to draw conclusions about the population, we need the variance of $\bar{y}$. It turns out that under simple random sampling,

(Proof: see section 2.9)

But, we generally don't know σ^2, so we'll use an unbiased estimator of $\text{Var}(\bar{y})$:

The square root of the variance of the estimator is its standard error (fun fact: standard errors are not usually unbiased).

This quantity

is called the **finite population correction factor**, or **fpc** for short.

Keep in mind that we've sampled $\frac{n}{N}$ of the population. This is called the **sampling fraction**. Why does the sampling fraction matter?

Example: For the baseball SRS

We know:

With the fpc:

```
var.salary<-var(baseballsrs$salary)
var.salary
```

```
[1] 8.804137e+12
```

```
se.mean<-sqrt((var.salary/n)*(1-(n/N)))
se.mean
```

```
[1] 218283.4
```

Without the fpc:

```
se.nofpc<-sqrt(var.salary/n)
se.nofpc
```

```
[1] 242269
```

What if we had sampled fewer players? Let's try 30 instead of 150. Now:

```
set.seed(98263476)
sampsize<-30
indexsmall<-srswor(sampsize,N)
srssmalln<-getdata(baseball,indexsmall)
meansmall<-mean(srssmalln$salary)
meansmall
```

```
[1] 1904472
```

```
var.small<-var(srssmalln$salary)
var.small
```

```
[1] 4.76297e+12
```

```
se.small<-sqrt((var.small/sampsize)*(1-(sampsize/N)))
se.small
```

```
[1] 390883
```

```
nofpc.small<-sqrt(var.small/sampsize)
nofpc.small
```

```
[1] 398454.1
```

We may also be interested in a measure of relative variability (this will be most helpful in the future for selecting sample size). The **coefficient of variation** (CV) of $\bar{y}$ is

The CV does not depend on the units of measurement. We can estimate the CV using

Example: For the baseball SRS, the estimated CV is

```
CV<-se.mean/mean.salary
CV
```

```
[1] 0.1094764
```

The salary data are quite skewed, so we should also report the median:

```
med.salary<-median(baseballsrs$salary)
med.salary
```

```
[1] 525000
```

3.2.2 Estimating a population total

As a reminder, the population total is

We can also express the population total as a function of the population mean

It's then natural to estimate t using

It's easy to see that

The CV doesn't change because the Ns cancel!

Example: Let's estimate the total number of runs across the 2004 season, based on our SRS of $n = 150$ players. First, let's start with the mean runs

```
mean.runs<-mean(baseballsrs$run)
mean.runs
```

```
[1] 24.39333
```

```
var.runs<-var(baseballsrs$run)
var.runs
```

```
[1] 998.2268
```

```
se.meanrun<-sqrt((var.runs/n)*(1-(n/N)))
se.meanrun
```

```
[1] 2.324299
```

We can now use these to find an estimate of the total number of runs, along with its standard error:

```
tot.runs<-N*mean.runs
tot.runs
```

```
[1] 19441.49
```

```
se.totrun<-N*se.meanrun
se.totrun
```

```
[1] 1852.466
```

Again, since we have the whole population we can calculate the true total number of runs:

```
pop.totrun<-N*mean(baseball$run)
pop.totrun
```

```
[1] 20515
```

3.2.3 Estimating a population proportion

A proportion is just a special case of a mean! Let p be the proportion of the population that has a characteristic, and define y_i as

Then,

and $\widehat{p}$ is an unbiased estimator for p.

Now let's consider the variance.

If we plug $\widehat{p}$ in for p:

Example: Let's estimate the population proportion of players who had at least one home run. First, we'll need to do some fiddling with the data set.

```
baseballsrs$hr<-rep(0,n)
baseballsrs$hr[baseballsrs$homerun>0]<-1
head(baseballsrs)
```

```
   ID_unit team leagueid   player  salary pos gplay gstart inning putout assist
8        8  ANA       AL figgich0  320000  3B   148     80   2116     57    129
10      10  ANA       AL greggke0  301500   P     5      0    263      2      5
21      21  ANA       AL rodrifr0  375000   P     3      0    252      5      9
31      31  ARI       NL cintral0  335000  SS   154    125   3297    141    383
33      33  ARI       NL daiglca0  300000   P    11     10    147      5      6
50      50  ARI       NL sexsori0 8725000  1B    23     23    613    199     27
   error dplay pball gbat atbat run hit secbase thirdbase homerun rbi stolenb
8     11     9    -9  148   577  83 171      22        17       5  60      34
10     0     1    -9    5     0   0   0       0         0       0   0       0
21     0     0    -9    3     0   0   0       0         0       0   0       0
31    15    61    -9  154   564  56 148      31         7       4  49       3
33     0     0    -9   11    17   2   2       2         0       0   0       0
50     1    19    -9   23    90  20  21       4         0       9  23       0
   csteal walk strikeout intwalk hbp sacrhit sacrfly gidplay hr
8      13   49        94       0   0      10       2       6  1
10      0    0         0       0   0       0       0       0  0
21      0    0         0       0   0       0       0       0  0
31      3   31        59       2   2      12       4      11  1
33      0    0         7       0   0       1       0       1  0
50      0   14        21       0   0       0       0       2  1
```

Now we can use our new variable to estimate the proportion of all players that had at least one homerun:

```
prop.hr<-mean(baseballsrs$hr)
prop.hr
```

```
[1] 0.5066667
```

and find its standard error

```
fpc<-1-(n/N)
fpc
```

```
[1] 0.8117942
```

```
var.prophr<-(fpc*prop.hr*(1-prop.hr))/(n-1)
var.prophr
```

```
[1] 0.001361829
```

```
se.prophr<-sqrt(var.prophr)
se.prophr
```

```
[1] 0.03690296
```

3.3 Sampling Weights

Earlier, we defined the **inclusion probability** (π_i) for each sampling unit. This is the probability that unit i is included in the sample. In probability samples, the inclusion probabilities are used to calculate $\bar{y}$ and $\hat{t}$.

To see how they are used, we need a new concept–the **sampling weight**.

Definition: Sampling weight

The **sampling weight**

For example, with the baseball data

The sampling weight is also sometimes called the **design weight**.

For simple random samples, all sampling units have the same inclusion probability:

This means that all sampling units also have the same sampling weight:

Because all units have the same sampling weight, an SRS is a **self-weighting sample**. However, just because a sample is self-weighting does not mean it's an SRS. It's possible to have a sample design where every unit has the same inclusion probability, but not all possible samples have the same chance to be selected.

As mentioned above, we can use the sampling weights w_i to calculate $\bar{y}$ and $\hat{t}$ (and, thus, $\hat{p}$).

For an SRS, $w_i = N/n$, so

We didn't need to use them in baseball example to get $\bar{y}$, $\hat{t}$, and $\hat{p}$. But, survey software packages in R and SAS are written to estimate the parameters for any probability sampling design so they require the data set include sampling weights.

Example: Let's add the sampling weights to the baseball data.

```
baseballsrs$sampwt<-rep(N/n,n)
head(baseballsrs)
```

```
   ID_unit team leagueid   player  salary pos gplay gstart inning putout assist
8        8  ANA       AL figgich0  320000  3B   148     80   2116     57    129
```

```
10      10  ANA       AL greggke0  301500   P     5      0    263      2      5
21      21  ANA       AL rodrifr0  375000   P     3      0    252      5      9
31      31  ARI       NL cintral0  335000  SS   154    125   3297    141    383
33      33  ARI       NL daiglca0  300000   P    11     10    147      5      6
50      50  ARI       NL sexsori0 8725000  1B    23     23    613    199     27
   error dplay pball gbat atbat run hit secbase thirdbase homerun rbi stolenb
8     11     9    -9  148   577  83 171      22        17       5  60      34
10     0     1    -9    5     0   0   0       0         0       0   0       0
21     0     0    -9    3     0   0   0       0         0       0   0       0
31    15    61    -9  154   564  56 148      31         7       4  49       3
33     0     0    -9   11    17   2   2       2         0       0   0       0
50     1    19    -9   23    90  20  21       4         0       9  23       0
   csteal walk strikeout intwalk hbp sacrhit sacrfly gidplay hr   sampwt
8      13   49        94       0   0      10       2       6  1 5.313333
10      0    0         0       0   0       0       0       0  0 5.313333
21      0    0         0       0   0       0       0       0  0 5.313333
31      3   31        59       2   2      12       4      11  1 5.313333
33      0    0         7       0   0       1       0       1  0 5.313333
50      0   14        21       0   0       0       0       2  1 5.313333
```

The sampling weight is $N/n = 797/150 = 5.31333$, indicating that player in our SRS represents 5.31333 players in the population of all 2004 MLB players.

We can then use the sampling weights to calculate the same estimates we did earlier, using a built in function in the `survey` package in R. We first have to define the survey design, using the `svydesign` function:

```
library(survey)
baseballdesign<-svydesign(id=~1, weights=~sampwt, fpc=rep(N,n), data=baseballsrs)
```

The `svydesign` function has several inputs:

- id
- weights
- fpc
- data

There are other inputs that we don't need right now, but we'll add them as we encounter more complicated sampling designs.

Let's look the information included in `baseballdesign`:

```
baseballdesign
```

```
Independent Sampling design
svydesign(id = ~1, weights = ~sampwt, fpc = rep(N, n), data = baseballsrs)
```

R tells us this is an Independent Sampling Design, which means there are no clusters or strata.

Now that we've set up the sampling design in R, we can use it calculate the same estimates we did before with a lot less code. We can use the functions `svymean` and `svytotal` to get $\bar{y}$ and $\hat{t}$, respectively, along with their standard errors. The generic code is:

```
ybar<-svymean(~var_interest, design)
```

```
that<-svytotal(~var_interest, design)
```

Example: For estimating average salary and total runs in 2004 we'd use:

```
salmean<-svymean(~salary,baseballdesign)
salmean
```

```
          mean     SE
salary 1993886 218283
```

```
runtotal<-svytotal(~run,baseballdesign)
runtotal
```

```
    total     SE
run 19442 1852.5
```

The same values we got before. Since $\hat{p}$ is really a mean, we can also use `svymean` to get $\hat{p}$ and its SE:

```
hrprop<-svymean(~hr,baseballdesign)
hrprop
```

```
      mean     SE
hr 0.50667 0.0369
```

Again, the same values we got before.

3.4 Confidence Intervals

Once we select the random sample and use the resulting data to construct estimates of a population value (mean, total, proportion), the next step is typically to make some statement about the accuracy of the estimate. This is most often accomplished by constructing a **confidence interval** for the true population value.

The confidence intervals we'll construct have the same interpretation as those we constructed in classes like STAT 218, 318, 102, 212, 380.

Definition: Confidence interval

A **confidence interval**

The expected proportion of intervals containing the true population value is the **coverage probability**. So, for example, an interval with a coverage probability of 0.95 means

But, there are differences. In probability sampling with a finite population, only a finite number of possible samples exist, and we know the probability with which each sample will be chosen. This means we *could* generate all possible samples, and all possible intervals. From these, we could then observe how many capture the population value, and get the *exact* coverage probability.

In real life, we can't do this

But, the confidence intervals we used in STAT 218/318/102/212/380 (which assumed an infinite population) relied on asymptotic theory–what happens as $n \to \infty$. For example, we know and love the Central Limit Theorem:

But we've only got a finite population! How can we use a result about $n \to \infty$ when we've got a finite population?

Happily, there is a version of the Central Limit Theorem for SRSWOR, due to Hájek (1960). If some conditions that are *way* too technical for us to talk about are true and if

then

This means we could use this confidence interval for the population mean:

Again, we don't know σ! We could substitute in s and use the standard error. And, in practice, we typically substitute $t_{\alpha/2,df=n-1}$ for $z_{\alpha/2}$. In small samples, it's more conservative (wider confidence intervals) and with large samples $t_{\alpha/2,n-1} \approx z_{\alpha/2}$ anyway. So, we'll use the interval

to calculate a $(1-\alpha) \times 100\%$ confidence interval for the from population mean from an SRS.

In general,

We have the same question come up that comes with the regular CLT: how large is large enough for n (and N?) It depends

Some rules of thumb:

- If the population is approximately normal,
-

Sugden et al. (2000) recommended the minimum sample size

Another way to check whether the distribution is too skewed to use the CLT is bootstrapping!

Example: Back to the baseball data. Let's get a 95% confidence interval for the mean salary of players in 2004. We saw already that based on our SRS of $n = 150$, $\bar{y} = 1993886$ and $\text{SE}(\bar{y}) = 218283.4$. We can use the `qt(alpha/2,df)` function to get the t multiplier in the confidence interval.

```
tcrit<-qt(0.025,n-1,lower.tail=FALSE)
lower<-mean.salary-tcrit*se.mean
upper<-mean.salary+tcrit*se.mean
lower
```

```
[1] 1562555
```

```
upper
```

```
[1] 2425217
```

Should we have even used the t-interval for the salary data? It's pretty skewed. We can find the skewness to check.

```
library(moments)
salskew<-skewness(baseballsrs$salary)
salskew
```

```
[1] 2.538835
```

And using the Sudgen et al. (2000) recommendation, we need at least

```
nmin<-28 + 25*(salskew)**2
nmin
```

```
[1] 189.142
```

What about a 95% confidence interval for the total number of runs in 2004? We already found $\hat{t} = 19441.49$ and $\text{SE}(\hat{t}) = 1852.4661651$.

```
lowerrun<-tot.runs-tcrit*se.totrun
upperrun<-tot.runs+tcrit*se.totrun
lowerrun
```

```
[1] 15780.99
```

```
upperrun
```

```
[1] 23101.98
```

Let's try a 95% confidence interval for the proportion players with at least one home run. We know $\hat{p} = 0.5066667$ and $\text{SE}(\hat{p}) = 0.036903$.

```
lowerhr<-prop.hr-tcrit*se.prophr
upperhr<-prop.hr+tcrit*se.prophr
lowerhr
```

```
[1] 0.4337459
```

```
upperhr
```

```
[1] 0.5795874
```

It turns out the `survey` package has confidence intervals built in to the `svymean` and `svytotal` functions. Once we have created an object from either function, we can pass that object into the `confint` function. Here's the syntax:

```
confint(object,df= , level= )
```

So we could use

```
salmean<-svymean(~salary, baseballdesign)
salmean
```

```
          mean     SE
salary 1993886 218283
```

```
confint(salmean,df=n-1, level=0.95)
```

```
         2.5 %  97.5 %
salary 1562555 2425217
```

For more practice:

1. Estimating the mean number of errors committed by a player in 2004
    - draw a new sample of size $n = 150$ (with no seed or pick your own seed, so everyone gets a different sample) from the baseball data set
    - use `svymean` to a 90% confidence interval for the mean number of errors committed by a player in 2004
2. Estimating the total number of pitchers in 2004
    - using the same sample you drew above, create a 0/1 variable that identifies whether each player in the sample is a pitcher (note that `svydesign` must be run again if a new variable is added to the data set)
    - use `svymean` to get an estimate for $\hat{p}$
    - use `svytotal` to estimate the total number of pitchers in 2004
3. Estimate something of your choice!

3.5 Sample Size

Often the first question that comes up when designing a survey is how many observations are needed. This is often a pretty complicated question, and involves balancing the objectives of the study with the available budget. You also must consider that most studies have multiple characteristics being measured on each observation unit.

If the researcher can narrow down the most important variables to focus on one or two, there are four basic steps:

1. Decide how much precision you need (basically, how much variability are you willing to live with in the estimates)
2. Find an equation involving the sample size n and the precision (i.e., the margin of error)
3. Estimate anything you need to and solve for n (usually an estimate of the expected variance)
4. Go back to square one when the n turns out to be enormous.

We'll go through these steps one by one.

1. Decide how much precision you need

2. Find an equation involving n and the precision

Let's start by assuming the fpc is neglible (i.e., we have an infinite population)

Now, let's assume $\frac{n_0}{N} < 1$ (why is this a reasonable assumption?)

So, the sample size we need is

We could also consider finding the sample size to achieve a specified relative precision, call it r. The relative precision measures the distance between the estimate and the true value, scaled by the true value.

Sometimes we can calculate the required n at this stage!

Example: Suppose we want to estimate the proportion of 2004 baseball players that had at least one error, and we want to be able to calculate a 95% confidence interval and have a margin of error of 0.04.

3. Estimate anything you need to and solve for n

This usually involves estimating (guessing) the expected variance. Why didn't need to do this in the baseball example above?

How can we obtain the expected variance?

Another option is to use the CV for a sample of size 1: $\sigma/\bar{y}_{\mathcal{U}}$

If there are no other options, guess. An educated guess may use the suspected distribution of the data.

4: Go back to square one when the n turns out to be enormous

Example: A botanical researchers wishes to design a survey to estimate the number of birch trees in a study area. The study area has been divided into 1000 plots. From previous experience, the variance in the number of trees per plot is about $\sigma^2 \approx 45$. What sample size should be used to estimate the total number of trees in the study area to within 500 trees of the true value with 95% confidence?

To within 1000 trees?

To within 2000 trees?

Does the finite population correction factor matter here? How do you know?

3.6 Systematic Sampling

Suppose we have a population with N units, and the units are numbered from 1 to N in some order. To select a sample of n units, we:

For example,

Note that

This type of sampling design is called a **systematic sample**

Why isn't this a SRS?

When will a systematic sample behave like an SRS?

When will a systematic sample not behave like an SRS?

There are a couple of advantages over the SRS:

3.7 Considerations for Choosing an SRS Design

When Shouldn't You Use an SRS?

- You really should be carrying out a designed experiment

- You don't have a sampling frame or it's too expensive

- You've got additional information

Why Should You Use an SRS?

- You don't have extra information

- Explainability is super important

- Easier computationally

4 Stratified Sampling

In the last chapter, we noted that an SRS isn't the best option if we have additional information that could help us come up with a more cost-effective sampling design. One such option is **stratified sampling**.

4.1 Overview of Stratified Sampling

Basic Concept: Stratification

The idea of **stratification** is

For example,

The basic set-up for stratified sampling is as follows:

-
-
-

What should we be thinking about to decide whether or not to stratify?

- Is there a chance we could draw a really bad sample?

- Do we want to draw conclusions about a particular subgroup of the population (especially with a particular precision)?

- Is it more convenient or cheaper?

- Can we construct the strata in a way that our estimates will be more precise?

If the answer to any of the above is yes, stratification is probably a good idea.

Example: In the last section, we selected a SRS of $n = 150$ baseball players from the 2004 MLB season. One of the variables we considered was salary. We saw that there was a lot of variability in the data, possibly due to the strong right skew. Is there also variability due to other factors? Maybe there is variability due to league as well.

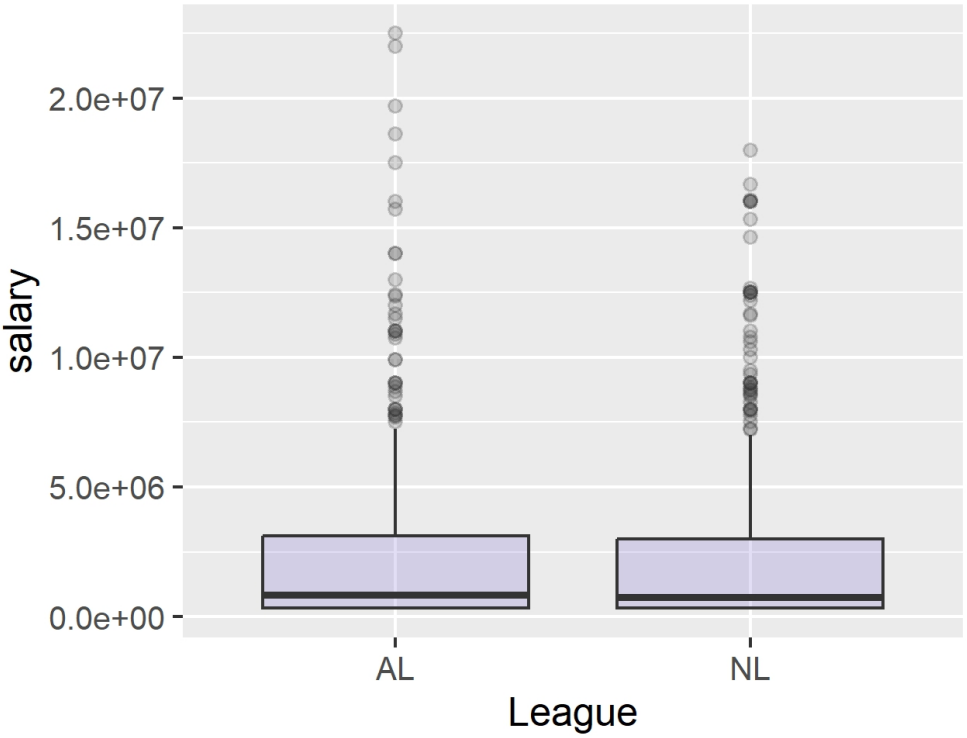


Figure 4.1: Baseball Salary by League

It looks like there is more variability among salary for the American League. Let's use League as our stratification variable. It meets the requirement:

We'll stick with a sample of $n = 150$ players. In the population, $372/797 = 0.46675$ players are in the AL and $425/797 = 0.53325$ players are in the NL. So, we'll sample 70 AL players and 80 NL players.

Another way to think about this …

When we found mean salary before, the SE was 218283.4. When I drew a stratified sample with the allocation above, the SE is 200945. So perhaps stratifying by league could be helpful in increasing precision.

4.2 How Does Stratified Sampling Work?

We're going to start by dividing the population (N sampling units) into H mutually exclusive and exhaustive strata,

-
-
-

Then, we select an independent probability sample from each stratum.

Some choices have to be made here, namely how to allocate the sample across the strata. In the baseball example, we used **proportional allocation**:

There are other methods of allocation as well, and we'll talk about these later.

What's important is that

We could use any sampling design (SRS, Systematic, etc.) within a stratum, and they don't all need to be the same. For simplicity, we'll assume a SRSWOR is selected within each stratum.

With this assumption of SRSWOR, we are carrying out **stratified random sampling**, and the biggest adjustment from SRS to stratified is really bookkeeping–there are a lot more subscripts to keep track of.

As a reminder, we assume

We'll start with new notation for the population quantities:

- N_h
- y_{hj}
- t_h
- t
- $\bar{y}_{h\mathcal{U}}$
- $\bar{y}_{\mathcal{U}}$
- $S_h^2 = \sigma_h^2$

The sample quantities will also have subscripts:

- $\mathcal{S}_h$
- n_h
- n
- $\bar{y}_h$
- $\hat{t}_h$
- s_h^2

4.3 Estimation with Stratified Random Samples

Suppose we sampled only the hth stratum:

This leads to an estimate for entire population total

and entire population mean

The estimated population mean is really just a weighted average of the sample stratum means, because

Because we took a SRS in each stratum, all the properties of SRS estimates are retained:

- **Unbiasedness**

- **Variance of the estimators**

- **Confidence intervals**

Example: In Homework 1, you worked with some data on the number of publications for a SRS of $n = 50$ faculty, out of a total of $N = 807$ faculty members. We'll use the areas biological sciences, physical sciences, social sciences, and humanities as the strata.

Stratum	Num of Faculty in Stratum	Num of Faculty in Sample
Biological sciences	102	7
Physical sciences	310	19
Social sciences	217	13
Humanities	178	11
Total	807	50

The actual publication data for each stratum is in the R code snippets below.

Let's set up a table similar to one shown in Example 3.3

Stratum	N_h	n_h	$\bar{y}_h$	s_h^2	$\hat{t}_h = N_h \bar{y}_h$	$(1 - \frac{n_h}{N_h}) N_h^2 \frac{s_h^2}{n_h}$
Bio sci	102	7	3.14	6.81		
Phy sci	310	19	2.105	8.21		
Soc sci	217	13	1.231	4.359		
Humanities	178	11	0.4545	0.8727		
Total	807	50				

So, our estimated total number of publications is

with standard error

So, a 95% confidence interval for the total number of publications is

In Chapter 3 of the notes, we observed that a proportion is just a mean where the only values the variable can take on are 0 and 1. We'll use the same idea here and define

Then we can estimate the overall proportion of the population with the characteristic as

with estimated variance

If we want to to estimate the total number of those in the population with the characteristic we can use

Example: Let's go back to the publications data and consider the proportion of faculty with at least one publication.

Stratum	N_h	n_h	N_h/N	$\bar{y}_h = \hat{p}_h$	$(1 - \frac{n_h}{N_h})(\frac{N_h}{N})^2(\frac{\hat{p}_h(1-\hat{p}_h)}{n_h-1})$
Bio sci	102	7	0.1264	0.857	
Phy sci	310	19	0.3841	0.474	
Soc sci	217	13	0.2689	0.308	
Humanities	178	11	0.2206	0.273	
Total	807	50			

So,

$\hat{p}_{str}$ =$\hat p_{str} =$

and

$\text{SE}(\hat{p}_{str}) =$

We can also estimate the total number of faculty with at least one publication and its standard error.

4.4 Sampling Weights

With simple random samples, we observed that the sampling weights were

because all sampling units have the same inclusion probability

and the sampling weights are

With stratified sampling, the inclusion probability may not be the same for all units (they will be the same for all units within the same stratum).

The **sampling weight for unit** j of stratum h is

The interpretation of the sampling weight is the same as it was with SRS!

Just as before, the sampling weight is the inverse of the inclusion probability:

We can again re-write the population estimates using the sampling weights

In the last section, we observed that simple random samples are **self-weighting** samples, because each sample unit has the same weight. Stratified random samples will be self-weighting samples if

Even if this is true, however, we still need to acknowledge the stratification in the estimation because it will make a difference in the variances.

We can use the `survey` package to calculate estimates using the `svymean` and `svytotal` functions, but we do need to specify the stratification variable and the sampling weights in the sampling design (i.e., in `svydesign`). Let's look more carefully at the R code for the publication data. We also have a new function, `svyby`.

First, we need to get the data in R. I am **not** an elegant R coder, I'm sure there are more efficient ways to do this.

```
biopubs<-c(rep(0,1),rep(1,2),rep(3,1),rep(5,2),rep(7,1))
biopubs
```

```
[1] 0 1 1 3 5 5 7
```

```
phypubs<-c(rep(0,10),rep(1,2),rep(3,1),rep(4,2),rep(5,1),rep(6,1),rep(8,2))
socpubs<-c(rep(0,9),rep(2,1),rep(4,2),rep(6,1))
humpubs<-c(rep(0,8),rep(1,2),rep(3,1))

biopubsdf<-data.frame(biopubs)
phypubsdf<-data.frame(phypubs)
socpubsdf<-data.frame(socpubs)
humpubsdf<-data.frame(humpubs)

biopubsdf$stratum<-"Biological"
biopubsdf$pubs<-biopubsdf$biopubs
biopubsdf$Nh<-102
biopubsdf$samp<-7
biopubsdf<-biopubsdf[,c(2,3,4,5)]

biopubsdf
```

```
     stratum pubs  Nh samp
1 Biological    0 102    7
2 Biological    1 102    7
3 Biological    1 102    7
4 Biological    3 102    7
5 Biological    5 102    7
6 Biological    5 102    7
7 Biological    7 102    7
```

```
phypubsdf$stratum<-"Physical"
phypubsdf$pubs<-phypubsdf$phypubs
phypubsdf$Nh<-310
phypubsdf$samp<-19
phypubsdf<-phypubsdf[,c(2,3,4,5)]
socpubsdf$stratum<-"Social"
socpubsdf$pubs<-socpubsdf$socpubs
socpubsdf$Nh<-217
```

```
socpubsdf$samp<-13
socpubsdf<-socpubsdf[,c(2,3,4,5)]
humpubsdf$stratum<-"Humanities"
humpubsdf$pubs<-humpubsdf$humpubs
humpubsdf$Nh<-178
humpubsdf$samp<-11
humpubsdf<-humpubsdf[,c(2,3,4,5)]

pubsdf<-rbind(biopubsdf,phypubsdf,socpubsdf,humpubsdf)
pubsdf
```

```
      stratum pubs  Nh samp
1  Biological    0 102    7
2  Biological    1 102    7
3  Biological    1 102    7
4  Biological    3 102    7
5  Biological    5 102    7
6  Biological    5 102    7
7  Biological    7 102    7
8    Physical    0 310   19
9    Physical    0 310   19
10   Physical    0 310   19
11   Physical    0 310   19
12   Physical    0 310   19
13   Physical    0 310   19
14   Physical    0 310   19
15   Physical    0 310   19
16   Physical    0 310   19
17   Physical    0 310   19
18   Physical    1 310   19
19   Physical    1 310   19
20   Physical    3 310   19
21   Physical    4 310   19
22   Physical    4 310   19
23   Physical    5 310   19
24   Physical    6 310   19
25   Physical    8 310   19
26   Physical    8 310   19
27     Social    0 217   13
28     Social    0 217   13
29     Social    0 217   13
30     Social    0 217   13
31     Social    0 217   13
```

```
32     Social    0 217   13
33     Social    0 217   13
34     Social    0 217   13
35     Social    0 217   13
36     Social    2 217   13
37     Social    4 217   13
38     Social    4 217   13
39     Social    6 217   13
40 Humanities    0 178   11
41 Humanities    0 178   11
42 Humanities    0 178   11
43 Humanities    0 178   11
44 Humanities    0 178   11
45 Humanities    0 178   11
46 Humanities    0 178   11
47 Humanities    0 178   11
48 Humanities    1 178   11
49 Humanities    1 178   11
50 Humanities    3 178   11
```

```
pubsdf$sampwt<-with(pubsdf,Nh/samp)
pubsdf
```

```
      stratum pubs  Nh samp   sampwt
1  Biological    0 102    7 14.57143
2  Biological    1 102    7 14.57143
3  Biological    1 102    7 14.57143
4  Biological    3 102    7 14.57143
5  Biological    5 102    7 14.57143
6  Biological    5 102    7 14.57143
7  Biological    7 102    7 14.57143
8    Physical    0 310   19 16.31579
9    Physical    0 310   19 16.31579
10   Physical    0 310   19 16.31579
11   Physical    0 310   19 16.31579
12   Physical    0 310   19 16.31579
13   Physical    0 310   19 16.31579
14   Physical    0 310   19 16.31579
15   Physical    0 310   19 16.31579
16   Physical    0 310   19 16.31579
17   Physical    0 310   19 16.31579
18   Physical    1 310   19 16.31579
19   Physical    1 310   19 16.31579
```

```
20   Physical    3 310   19 16.31579
21   Physical    4 310   19 16.31579
22   Physical    4 310   19 16.31579
23   Physical    5 310   19 16.31579
24   Physical    6 310   19 16.31579
25   Physical    8 310   19 16.31579
26   Physical    8 310   19 16.31579
27     Social    0 217   13 16.69231
28     Social    0 217   13 16.69231
29     Social    0 217   13 16.69231
30     Social    0 217   13 16.69231
31     Social    0 217   13 16.69231
32     Social    0 217   13 16.69231
33     Social    0 217   13 16.69231
34     Social    0 217   13 16.69231
35     Social    0 217   13 16.69231
36     Social    2 217   13 16.69231
37     Social    4 217   13 16.69231
38     Social    4 217   13 16.69231
39     Social    6 217   13 16.69231
40 Humanities    0 178   11 16.18182
41 Humanities    0 178   11 16.18182
42 Humanities    0 178   11 16.18182
43 Humanities    0 178   11 16.18182
44 Humanities    0 178   11 16.18182
45 Humanities    0 178   11 16.18182
46 Humanities    0 178   11 16.18182
47 Humanities    0 178   11 16.18182
48 Humanities    1 178   11 16.18182
49 Humanities    1 178   11 16.18182
50 Humanities    3 178   11 16.18182
```

With data in place, we can now find some estimates for the whole population of 807 faculty members:

```
library(survey)

pubsdes<-svydesign(id=~1,strata=~stratum,weights=~sampwt,fpc=~Nh,data=pubsdf)
strmean<-tapply(pubsdf$pubs,pubsdf$stratum,mean)
strmean
```

```
Biological Humanities   Physical     Social
 3.1428571  0.4545455  2.1052632  1.2307692
```

```
strvar<-tapply(pubsdf$pubs,pubsdf$stratum,var)
strvar
```

```
Biological Humanities   Physical     Social
 6.8095238  0.8727273  8.2105263  4.3589744
```

```
pubsstrat<-svyby(~pubs,by=~stratum,pubsdes,svymean, keep.var=TRUE)
confint(pubsstrat,df=50-4)
```

```
                2.5 %   97.5 %
Biological  1.22687120 5.058843
Humanities -0.09463164 1.003723
Physical    0.82324081 3.387286
Social      0.10064302 2.360895
```

```
svytotal(~pubs,pubsdes)
```

```
      total     SE
pubs 1321.2 256.15
```

```
pubsdf$yes<-rep(0,50)
pubsdf$yes[pubsdf$pubs>0] <- 1
pubsdes<-svydesign(id=~1,strata=~stratum,weights=~sampwt,fpc=~Nh,data=pubsdf)
svymean(~yes,pubsdes)
```

```
       mean     SE
yes 0.43319 0.0658
```

```
svytotal(~yes,pubsdes)
```

```
     total     SE
yes 349.59 53.128
```

4.5 Sample Size and Allocation

With simple random samples, we only had one design feature we needed to determine: the sample size n. With stratified sampling, we need to determine:

It's actually easier to look at these in reverse order. We'll start by assuming the strata have already been defined, and consider how to allocate observations among them.

4.5.1 Allocation in Stratified Random Sampling

The general set-up is

There are several options for determining how many observations to sample in each stratum. Some allocation strategies deal with the whole population, others deal with subpopulations. Yet others are a compromise between the two. The best allocation strategy depends on the goals of the study.

4.5.1.1 Whole Population Allocation

With whole population allocation, we first determine how many sampling units we can collect/afford in the entire sample. That is, we first determine n. Then, we turn attention to how to divide n across the strata. Whole population allocation is best when the objective of the study is to estimate a characteristic for the entire population. There are several factors that affect allocation rules

The simplest whole population method is **proportional allocation**.

Proportion Allocation

In this situation, the inclusion probability for unit j in stratum h is

and is the same across all strata. This also implies

One note: rounding in proportional allocation can lead to unequal (but approximately equal) sampling weights.

Example: With the baseball data, let's consider stratifying by position:

Stratum h	Stratum Total N_h	Stratum Sample Size n_h	Sample Weight w_{hj}
1 (1B)	66		
2 (2B)	50		
3 (3B)	53		
4 (C)	61		
5 (CF)	43		
6 (LF)	49		
7 (P)	376		
8 (RF)	54		
9 (SS)	45		
Total	797	150	

When the strata are large enough,

Proportional allocation tends to provide optimal estimates (optimal in the sense of minimum variance) when

Proportional allocation is a good choice if

If the variances are not approximately equal across strata, then **Neyman allocation** is an option.

Neyman Allocation

Suppose the within-stratum variances for y_{hj} (σ_h^2) vary across strata. To obtain a similar level of precision in each stratum, we'd need a higher sample size in strata that are more variable.
For example, consider the estimate of stratum total, $\hat{t}_h$. We saw earlier that

Neyman allocation suggests allocating stratum sample size in proportion to

It turns out (proof not shown) that the variance of an estimate for population mean or total (for a fixed total n) is minimized if

and we can see the logic of this by equating

Tiny Example: A population consists of three strata of sizes $N_1 = 150$, $N_2 = 90$, and $N_3 = 120$, so $N = 360$. Based on previous studies, we expect the standard deviations within each strata to be $\sigma_1 \approx 100$, $\sigma_2 \approx 200$, and $\sigma_3 \approx 300$. We know we can sample a total of $n = 12$ sampling units.

Proportional allocation:

Neyman allocation:

Neyman allocation tends to provide optimal (in the sense of minimum variance) estimates when

Considerations for selecting Neyman allocation:

Neyman allocation is a special case of **optimal allocation**, which considers both variability and cost, and is often used when the sampling units themselves are of different sizes. Larger units are likely to be more variable, and are sampled at a higher rate.

Examples: Sampling units that vary in size could be

To build in variable cost, suppose data collection costs c_h vary across strata.

Optimal allocation

In **optimal allocation**, the sample size is allocated to strata in proportion to

So again, we're going to equate

and get the allocation rule

With optimal allocation we see sample size for stratum h (n_h) is larger in strata where:

Example: We're designing a study where the objective is to estimate the fraction of UNL students on academic probation that have been attending academic coaching sessions. Here's our sample design

- **Frame**:

- **Stratum 1**: Students meeting with their advisors (based on notes in Student Success Hub)
  - $N_1 = 750$, $p_1 = 0.90$, $c_1 = \$100$
- **Stratum 2**: Students not meeting with their advisors (based on no notes in Student Success Hub)
  - $N_2 = 100$, $p_2 = 0.35$, $c_2 = \$400$
- **Sample size**: $n = 100$

Here's how optimal allocation would work:

Stratum h	N_h	$\sigma_h^2 \approx p_h(1-p_h)$	c_h	$N_h\sigma_h/\sqrt{c_h}$	$\frac{N_h\sigma_h/\sqrt{c_h}}{\sum_{h=1}^{H} N_h\sigma_h/\sqrt{c_h}}$	n_h
1: Notes	750		100			
2: No Notes	100		400			
Total	850					$n = 100$

To see a summary of whole population allocation strategies, let's again think about equating

and look at what terms cancel out in particular circumstances. Then we can make the following summary

4.5.1.2 Subpopulation Allocation

With subpopulation allocation, our main interest is in estimating stratum characteristics

One warning: **subpopulation allocation generally leads to less precise whole population estimates**.

If stratum costs and variances are equal, then **equal allocation** is an option. With equal allocation we assume

Equal allocation

The equal allocation rule is to divide n equally across the H strata. So,

Example:

Allocation for specific precision

If our real focus is on estimating one (or two) specific subpopulation values and σ_h^2 varies across strata, we can

Then, once each stratum sample size is determined, sum to get the total sample size:

This made lead to sample sizes that exceed the allowable budget. Some options for adjusting sample sizes when initial calculations look too high:

- Relax specifications for desired precision and recalculate stratum sample sizes. This can involve widening the allowable margin of error or decreasing the confidence level
- Decide the sample size you can afford and reduce stratum sample sizes proportionally (recheck the margin of error to make sure the confidence interval would still be useful)
- It's probably going to take some fiddling

4.5.1.3 Compromise Allocation

Suppose the objective of the study is to get reasonable estimates of both whole and subpopulation values. There are allocation rules that fall somewhere between optimal for stratum estimates and optimal for population estimates. It's easiest to visualize with a graph.

Assuming equal variances and cost

The compromise is **square root allocation**.

Square root allocation

It's a compromise in that

- More sampling units are allocated to small strata than would be with proportional allocation
- More sampling units are allocated to large strata than would be with equal allocation

Example: Let's compare the allocations among the baseball positions (and use $n = 153$, just so equal allocation gives us an integer)

Stratum h	Stratum Total N_h	$\sqrt{N_h}$	Proportional n_h	Equal n_h	Square root n_h
1 (1B)	66	8.124		17	
2 (2B)	50	7.071		17	
3 (3B)	53	7.280		17	
4 (C)	61	7.810		17	
5 (CF)	43	6.557		17	
6 (LF)	49	7		17	
7 (P)	376	19.391		17	
8 (RF)	54	7.348		17	
9 (SS)	45	6.708		17	
Total	797	77.289		153	

There are other strategies as well! All of these are just guidelines, and you can mix features of different allocation rules.

Example: Say you have 5 strata and can afford a total of $n = 4000$ observations. You may allocate 400 to each strata to achieve a margin of error of 5% percentage points on strata confidence intervals for proportions. You've only allocated $5 \times 400 = 2000$ observations, so you can 'spend' the remaining 2000 proportionally or optimally.

There are methods for determining total sample size as well, and those are derived similarly to what we did with simple random samples. They are complicated by having multiple variances for the multiple strata. More details may be found on page 96.

4.5.2 Defining Strata

We started our discussion of selecting sampling design elements by stating that it's easier to work in reverse order, and imagine that we already have defined strata. Now we turn our attention to actually selecting the strata.

First, an important note: stratification almost never hurts, even if we stratify poorly.

However, stratification will be most efficient when

This means

In general, more information = more strata.

Example: (3.13) The point in time count of people experiencing homelessness is carried out on a single night on one of the last 10 nights in January. Large cities mean large areas to visit. Here's how NYC does it:

4.5.3 Allocation Examples

For all examples, we'll be keeping in mind the general allocation rule

$$n_h = \frac{nN_h s_h/\sqrt{c_h}}{\sum_{i=1}^{H} N_h s_h/\sqrt{c_h}}$$

6/24. Suppose that a city has 90,000 dwelling units, of which 35,000 are houses, 45,000 are apartments, and 10,000 are condominiums.

(a) You believe that the mean electricity usage is about twice as much for houses as for apartments or condominiums, and that the standard deviation is proportional to the mean so that $S_1 = 2S_2 = 2S_3$. How would you allocate a stratified sample of 900 observations if you wanted to estimate the mean electricity consumption for all households in the city?

(b) Now suppose that you take a stratified random sample with proportional allocation and want to estimate the overall proportion of households in which energy conservation is practiced. If 45% of house dwellers, 25% of apartment dwellers, and 3% of condominium residents practice energy conservation, what is p for the population? What gain would the stratified sample with proportional allocation offer over an SRS? That is, what is $V_{\text{prop}}(\hat{p}_{\text{str}})/V_{\text{SRS}}(\hat{p}_{\text{SRS}})$?

8/32. A public opinion researcher has a budget of $150,000 for interviewing households in a survey. She knows that 90% of all households have telephones. Telephone interviews cost $100 each; in-person interviews cost $300 each if all interviews are conducted in person, and $400 each if only nonphone households are interviewed in person (because there will be extra travel costs). Assume that the variances in the phone and nonphone groups are similar. How many households should be interviewed in each group if

(a) All households are interviewed in person.

(b) Households with a phone are contacted by telephone and households without a phone are contacted in person.

17/36. Lydersen and Ryg (1991) used stratification techniques to estimate ringed seal populations in Svalbard fjords. The 200 km^2 study area was divided in to three zones: Zone 1, outer Sassenfjorden, was covered with relatively new ice during the study period in March, 1990, and had little snow cover; Zone 3, Tempelfjorden, had a stable ice cover throughout the year; Zone 2, inner Sassenfjorden, was intermediate between the stable Zone 3 and unstable Zone 1. Ringed seals need good ice to establish territories with breathing holes, and snow cover enables females to dig out birth lairs. Thus, it was thought that the three zones would have different seal densities. The investigators took a stratified random sample of 20% of the 200 1-km^2. The following table gives the number of plots, and the number of plots sampled, in each zone, along with the sample mean (and variance) number of breathing holes found in each zone:

Zone	Number of Plots	Plots Sampled	$\bar{y}_h$	s_h^2
1	68	17	1.765	3.316
2	84	12	4.417	11.538
3	48	11	10.545	46.073
Total	200	40		

(a) Estimate the total number of breathing holes in the study region, along with its standard error.

(b) If you were designing this survey, how would you allocate observations to strata if the goal is to estimate the total number of breathing holes? If the goal is to compare the density of breathing holes in the three zones (want equal precision for $\bar{y}_h$ in the three zones)?

(c) The data are in the file `seals.csv` in Canvas. Use R to verify your answer in (a).

27/70. Suppose that, instead of emailing all 15,769 persons in the current membership list of the ASA as described in Exercise 13 of Chapter 1, the researchers wanted to take a stratified random sample of 1,100 of the 5,803 female and 9,966 male ASA members. What should the same sizes for men and women be if:

(a) Proportional allocation is used?

(b) Neyman allocation is used, and it is assumed that $S^2_{\text{women}} = (11/5) \times S^2_{\text{men}}$?

33/33. A stratified sample is being designed to estimate the prevalence p of a rare characteristic, say the proportion of residents of Milwaukee, Wisconsin, who have Lyme design. Stratum 1, with N_1 units, has a high prevalence of the characteristic; stratum 2, with N_2 units, has low prevalence. Assume that the cost to sample a unit (for example, the cost to select a person for the sample and determine they have Lyme disease) is the same for each stratum, and that at most 2000 persons are to be sampled.

(a) Let p_1 and p_2 be the proportions in stratum 1 and stratum 2 with the rare characteristic. If $p_1 = 0.10, p_2 = 0.03$, and $N_1/N = 0.4$, what are n_1 and n_2 under optimal allocation?

(b) If $p_1 = 0.10, p_2 = 0.03$, and $N_1/N = 0.4$, what is $V(\widehat{p}_{\text{str}})$ under proportional allocation? Under optimal allocation? What is the variance if you take an SRS of 2000 units from the population?

5 Ratio and Regression Estimation

At this point, we are going to change focus for a bit. Rather than discussing different sampling designs, we're going to think about making the best use of survey data. Specifically, we're going to consider the case where in addition to the characteristic of interest (y), one or more additional variables related to y are also available on that sampling unit. These additional variables are denoted by x and are called **auxiliary variables**.

Examples:

Now, such auxiliary information can be used in setting up the sampling design.

However, we are now concerned with using it for the estimation of population characteristics. The general idea is that we will exploit the relationship between y and x to get more precise estimates than we could get from y alone. We'll consider two methods for using this auxiliary information: ratio estimators and regression estimators.

In both cases, we may know the xs for the whole population or we may know the xs only for the sample.

Examples:

We're going to start with ratio estimators, and we're going to assume the traditional, design-based context. That is, we're going to assume:

5.1 Ratio Estimation in the SRS

Here's the set-up we're going to start with:

This means we're assuming

There are two potential goals of ratio estimation.

1.

2.

We'll need some new notation.

- Sampling plan:
- $t_y =$
- $\bar{y}_{\mathcal{U}} =$
- $\bar{y} =$
- $t_x =$
- $\bar{x}_{\mathcal{U}} =$
- $\bar{x} =$
- $B =$
- $\hat{B} =$

From these, we can get some pretty natural estimators for population characteristics:

Ratio estimators (and regression estimators) work best when there is a strong relationship between y and x. We're assuming a linear relationship between them, so

Example:

Before we jump into the properties of these estimators (bias, variance), let's explore why a ratio estimator may be useful.

5.1.1 Why consider a ratio estimator?

Maybe we actually want to estimate a ratio.

Suppose we want to estimate average income per adult in the population. We record y_i = household income and x_i = number of adults living in a household.

The key question to ask is: if you took a different sample, would the denominator change? If yes, then you're using a ratio estimator.

Maybe we want to estimate a population total, but we don't know N.

Up to now, we've used $\hat{t} = N\bar{y}$ to estimate a population total. But what if we don't know N?

Example: Suppose we want to estimate the amount of juice that will be produced by a shipment of oranges. We could take a sample of n oranges and record $y_1, \ldots, y_n$ (the amount of juice produced by each orange in the sample), and find the average amount of juice per orange, $\bar{y}$. But, to estimate the total amount of juice we'd need the total number of oranges in the shipment. What's easy to measure?

Maybe we want to increase the precision of $\bar{y}$ and $\hat{t}$.

This is one of the primary uses of ratio estimation–to get better estimates of the quantities we've already been focused on.

Example: A manager at a mill wants to estimate the total weight of dry wood (t_y) for a set number of truckloads of 5-foot bundles of pulp wood (freshly cut wood). The manager plans to:

1. Weigh the total amount of pulp wood (t_x)
2. Randomly select a sample of n bundles
3. Record the weight of the pulp wood (x_i) for each of the n bundles
4. Remove the bark and dry the wood from the n bundles
5. Record the weight of the dry wood (y_i) for each of the n bundles

Maybe we need to adjust estimates from the sample to reflect demographic totals.

We can use **poststratification** to adjust our estimates of population means and totals using information we know about the population.

Example: Suppose a small town of $N = 4000$ registered voters wants to estimate the total number of people that plan to vote early. They can use this information to figure out how many poll workers they need for election day. The town records show that of the 4000 registered voters, 2100 are registered Democrats and 1900 are registered Republicans. They take a random sample of $n = 200$ voters, and find that 39/85 registered Republicans plan to vote early and 75/115 registered Democrats plan to vote early.

Ratio estimation is used within each party, and we adjust our estimate accordingly. In this case:

Maybe we're hoping to account for nonresponse.

Not everyone responds to surveys and sometimes it's impossible to get data from each unit selected in our sample. But, if there's a relationship between y and x we can use that information to adjust our estimates.

Example: Suppose we want to estimate the total amount of money UNL students spent on books in Fall 2026. We distribute a survey to randomly selected enrolled students and record y_i = amount spent and x_i = number of credit hours enrolled.

We know $t_x =$

So, there are many reasons why we may want to use a ratio estimator. However, if we want to use them we need to know their properties so we can understand their behavior and calculate quantities like confidence intervals.

5.1.2 Ratio estimators and their properties

Unlike $\bar{y}$ or $\hat{t} = N\bar{y}$ in a SRS or stratified random sample, ratio estimators are usually biased for estimating $\bar{y}_{\mathcal{U}}$ and t_y. This is bias inherent to the estimation method, not selection bias.

Example: Suppose we want to estimate the total number of fish caught on a given day in nets at $N = 4$ sites. The entire population is

i: site	1	2	3	4
x_i: nets	4	5	8	5
y_i: fish	200	300	500	400

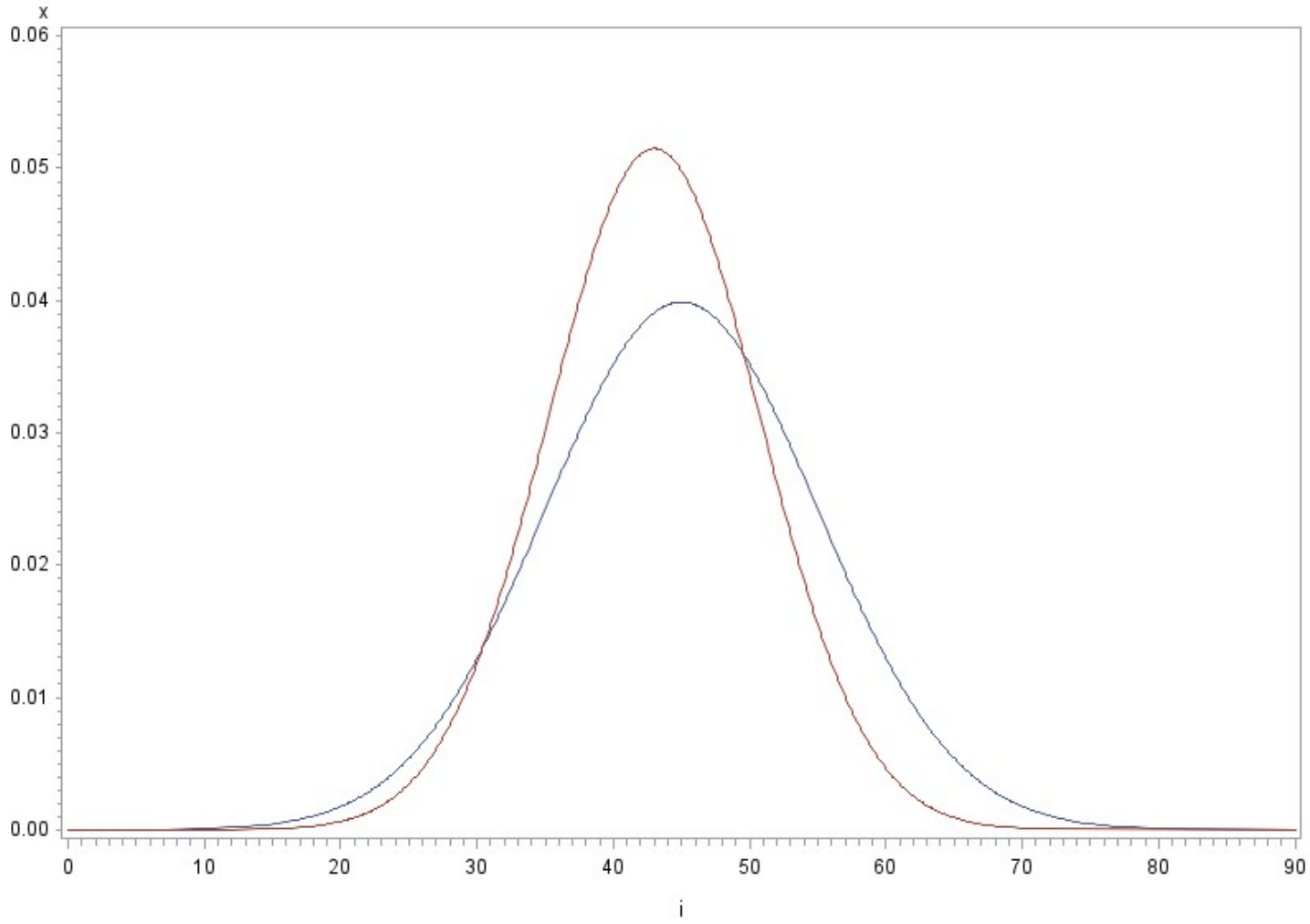
We will take a SRS of $n = 2$ sites and use ratio estimation to estimate the total number of fish caught. We know:

- $t_y =$
- $t_x =$
- $\binom{N}{n} = \binom{4}{2} =$

In this super small example, we can list all the possible samples.

Sample	Units	$\bar{x}_{\mathcal{S}}$	$\bar{y}_{\mathcal{S}}$	$\hat{B}$	$\hat{t}_{yr} = \hat{B}t_x$	$\hat{t}_{SRS}$
1	1,2					
2	1,3					
3	1,4					
4	2,3					
5	2,4					
6	3,4					

So if the ratio estimator is biased, why do we use it? We hope the reduced variance will compensate for the bias. That is, if x and y are highly (positively) correlated, we expect that the sampling distributions of $\bar{y}$ and $\hat{\bar{y}}_r$ will look like this compared to each other.



We have a quantity that summarizes both bias and variance!

Example:

Sample	Units	$\hat{t}_{yr}$	$[\hat{t}_{yr} - \mathrm{E}(\hat{t}_{yr})]^2$	$\hat{t}_{SRS}$	$[\hat{t}_{SRS} - \mathrm{E}(\hat{t}_{SRS})]^2$
1	1,2	1222		1000	
2	1,3	1283		1400	
3	1,4	1467		1200	
4	2,3	1354		1600	
5	2,4	1540		1400	
6	3,4	1523		1800	

Why is the bias there? Let's look at $\mathrm{E}[\hat{\bar{y}}_r]$:

Can we be more specific about the bias? Let's consider the bias of $\widehat{\bar{y}}_r$:

From this, we could show:

It can also be shown that:

What does all this mean? The take home message is the bias of $\hat{\bar{y}}_r$ (and $\hat{t}_{yr}$ and $\hat{B}$) is small if:

-
-
-
-
-

If the bias is small, we can approximate both the variance and the MSE using

Again, what does all this mean? The bottom line is the MSE and variance of $\hat{\bar{y}}_r$ (and $\hat{t}_{yr}$ and $\hat{B}$) is small if:

- n is large
- sampling fraction $\frac{n}{N}$ is large
- correlation R is close to 1
-

This approximation for the variance/MSE is still full of quantities we don't know! How can we get estimated variances based on the data we observed so we can calculate confidence intervals? Let's start with $\widehat{B}$:

From this, we can see what we'll use for $\widehat{\bar{y}}_r$ and $\widehat{t}_{yr}$:

We can then use these variances as we have before: take the square root to get the SE and find confidence intervals:

Example: Let's look at the orange juice example we considered earlier. The total shipment of oranges weighed 2000 pounds. 15 oranges were randomly selected. Let y_i = the weight of the orange and let x_i = the weight of the juice extracted from the orange.

Orange	y	x	$y - \hat{B}x$	$(y - \hat{B}x)^2$
1	0.16	0.22	0.014866	0.000220998
2	0.15	0.26	-0.021522	0.000463196
3	0.2	0.31	-0.004507	2.0313×10^{-5}
4	0.25	0.37	0.005911	3.49399×10^{-5}
5	0.16	0.28	-0.024716	0.000610881
6	0.27	0.38	0.019314	0.000373031
7	0.28	0.4	0.01612	0.000259854
8	0.16	0.21	0.021463	0.00046066
9	0.11	0.18	-0.008746	7.64925×10^{-5}
10	0.16	0.29	-0.031313	0.000980504
11	0.17	0.26	-0.001522	2.31648×10^{-6}
12	0.24	0.32	0.028896	0.000834979
13	0.21	0.33	-0.007701	5.93054×10^{-5}
14	0.11	0.16	0.004448	1.97847×10^{-5}
15	0.22	0.35	-0.010895	0.000118701
Total	2.85	4.32		0.00454

5.1.3 Sampling weights

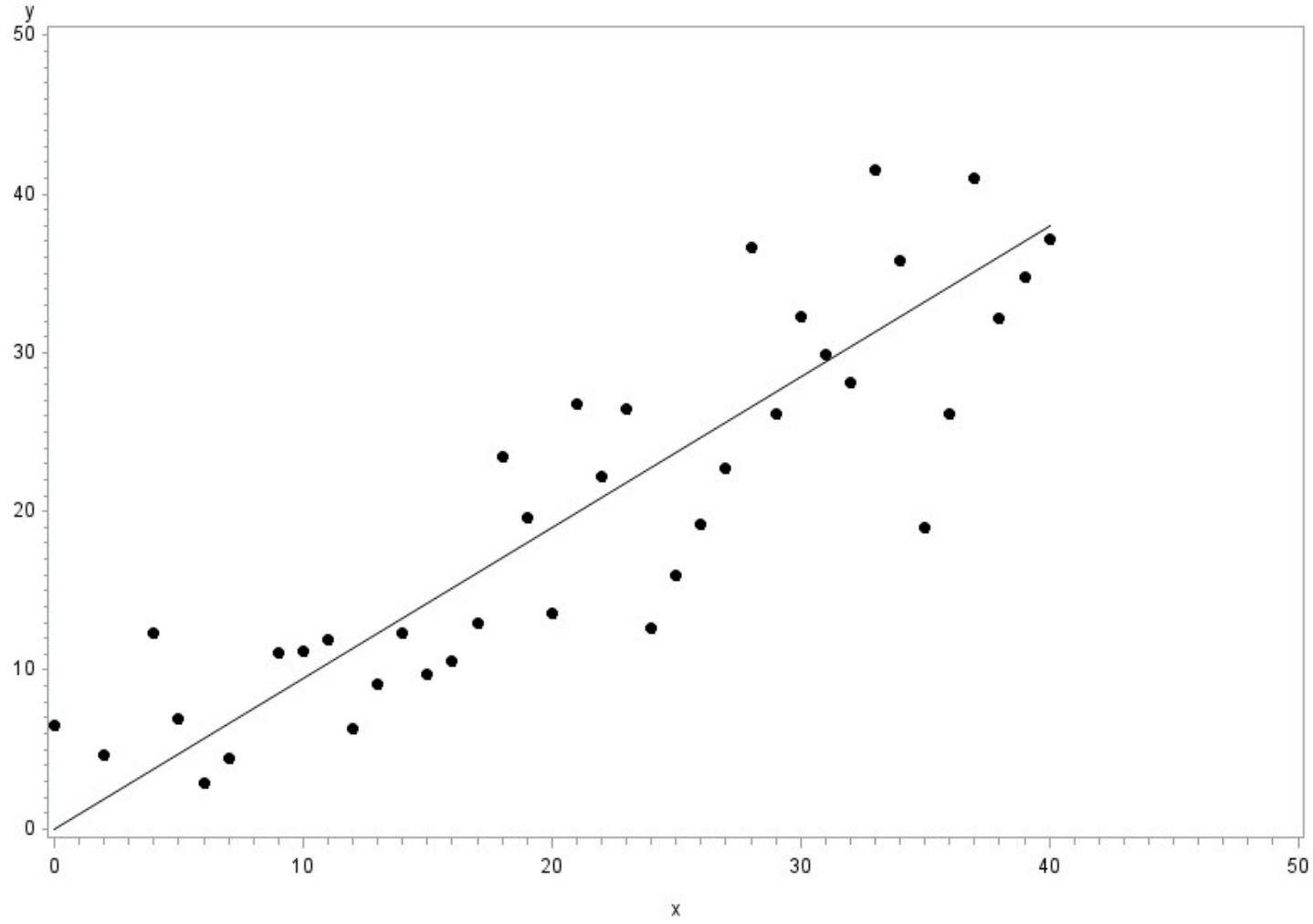
Recall the sampling weight is $w_i = 1/\pi_i$, where π_i is the inclusion probability. In previous chapters (SRS and Stratified) we observed that we we could use the sampling weights to get our estimates of population (or subpopulation) quantities. For example,

But, we also know

So, these $\bar{x}_{\mathcal{U}}/\bar{x}$ (or $t_x/\widehat{t}_x$) values serve as an adjustment to each sampling weight. Why do the weights need adjusting?

We're assuming

Let's add some data to the plot.



If $\bar{x} < \bar{x}_{\mathcal{U}}$

If $\bar{x} > \bar{x}_{\mathcal{U}}$

So, back to $t_x/\hat{t}_x$ as an adjustment to the sampling weights. Call this adjustment g_i:

We now have adjusted weights $w_i^* = w_i g_i$ and we can think about our estimator as

What makes these different from the sampling weights we saw earlier?

This means the sampling weights no longer sum to N, which they have done in previous sections. Instead, they are calibrated to the population total of the xs, t_x.

Example: Let's go back to our small example.

i: site	1	2	3	4
x_i: nets	4	5	8	5
y_i: fish	200	300	500	400

5.1.4 Analysis in R

We'll still need to set up a design, using the `svydesign` function. Since we're still assuming a SRS it will look just like it did before:

```
design<-svydesign(id=~1, weights=~sampwt, fpc=rep(N,n), data=dataset)
```

These are the unadjusted sampling weights, just the inverse of the inclusion probability. If we don't know N, we can set the sampling weights to all be 1 and omit the fpc.

There's a new function, `svyratio`, that will estimate B using $\hat{B} = \bar{y}/\bar{x}$ along with its standard error. It has the syntax:

```
est_ratio<-svyratio(numerator = ~y, denominator=~x, design=design)
```

We could also get a confidence interval for the ratio using the `confint` function. Once we have the estimated ratio, we can use the `predict` function to get the ratio estimates of $\bar{y}_U$ and t. We will need to specify t_x, the population total of the xs. We use the syntax:

```
xtotal<-value
predict(est_ratio,total=xtotal)
predict(est_ratio,total=xtotal/N)
```

Example: Back on Homework 1, you worked with the 'counties' data that includes information on land area, population, number of physicians, and other characteristics for a SRS of 100 counties (out of the 3141 counties) in the US. The data was collected in 1993 and at that time the total land area was 3536278 square miles and total population was 255077536.

Let's start by considering the characteristic of interest y = number of physicians and use x = population as an auxiliary variable.

We'll start by looking at the SRS estimate so we can compare.

```
counties<-read.csv("counties.csv",header=TRUE)
N<-3141
n<-nrow(counties)
counties$sampwt<-rep(N/n,n)

library(survey)
countiesdesign<-svydesign(id=~1, weights=~sampwt, fpc=rep(N,n), data=counties)

#SRS estimate#
totphys<-svytotal(~physician,countiesdesign)
totphys
```

```
           total     SE
physician 933411 491983
```

Now we can add our auxiliary variable. We'll first check the correlation between y and x:

```
cor(counties$physician, counties$totpop)
```

```
[1] 0.9771175
```

This is high! Population should be a good auxiliary variable. We'll estimate the ratio

```
Bhatphys<-svyratio(numerator=~physician,denominator=~totpop,design=countiesdesign)
Bhatphys
```

```
Ratio estimator: svyratio.survey.design2(numerator = ~physician, denominator = ~totpop,
    design = countiesdesign)
Ratios=
               totpop
physician 0.002507104
SEs=
                totpop
physician 0.0003445433
```

and use that to estimate the total number of physicians ($\hat{t}_y$)

```
xtotal<-255077536
predict(Bhatphys,total=xtotal)
```

```
$total
          totpop
physician 639506

$se
            totpop
physician 87885.27
```

Look at the reduction in the SE!

Now let's consider the characteristic of interest y = farm population and use x = land area as an auxiliary variable. Again, we'll start with the SRS estimate.

```
totfarmpop<-svytotal(~farmpop,countiesdesign)
totfarmpop
```

```
          total     SE
farmpop 3602319 329787
```

We'll compare it to the ratio estimate.

```
Bhatfarmpop<-svyratio(numerator=~farmpop,denominator=~landarea,design=countiesdesign)
Bhatfarmpop
```

```
Ratio estimator: svyratio.survey.design2(numerator = ~farmpop, denominator = ~landarea,
    design = countiesdesign)
Ratios=
        landarea
farmpop 1.213722
SEs=
         landarea
farmpop 0.1891048
```

```
landtotal<-3536278
predict(Bhatfarmpop,total=landtotal)
```

```
$total
        landarea
farmpop  4292058

$se
        landarea
farmpop 668727.3
```

The SE more than doubled! What's going on?

```
cor(counties$farmpop,counties$landarea)
```

```
[1] -0.05811006
```

5.2 Regression Estimation

With ratio estimation, we assumed a proportional relationship between our characteristic of interest y and an auxiliary variable x:

In the last section, we noted that assuming this relationship is equivalent to regression through the origin. Sometimes the relationship between x and y, even if strong, is not well represented by this assumption. Instead, perhaps a straight line with an intercept is a better depiction of the relationship:

Example: Suppose we want to estimate total yield of a crop in a field with $N = 100$ plots. We randomly select $n = 4$ plots using a SRS and in each plot we measure y_i = yield. Also available is the amount of fertilizer applied to the plot. This information is available on all 100 plots.

The new estimator we'll use is called a **regression** estimator, and we'll focus on estimating $\bar{y}_{\mathcal{U}}$. From $\bar{y}_{\mathcal{U}}$, we can extend to estimating t_y. Estimating $\bar{y}_{\mathcal{U}}$ turns out to be a pretty simple operation.

Assume the relationship between x and y follows

Let $\widehat{B}_0$ and $\widehat{B}_1$ be the usual ordinary least squares estimates of the intercept and slope (the typical regression estimators):

Before, with the ratio estimator

Now,

Again, the resulting regression estimator is an adjustment to the basic SRS estimator, and the adjustment is serving the same purpose it did with the ratio estimator.

Like the ratio estimator, the regression estimator is biased. But wait? Aren't the OLS estimators unbiased? What's going on?

With the ratio estimator, we showed

With the regression estimator,

Like the ratio estimator, the bias in the regression estimator is negligible with large samples.

Also like the ratio estimator, our estimated variance for $\hat{\bar{y}}_{reg}$ will be based on the variance of the residuals. But now,

You will sometimes also see the variance estimated using

This makes a bigger difference with small samples.

We can use $\widehat{\bar{y}}_{reg}$ to estimate a population total. Recall that

So, it should be intuitive to use

Example: Let's go back to the example we considered at the beginning of this section. We want to estimate total crop yield in a field with $N = 100$ plots. We use a SRS to randomly select $n = 4$ plots and measure y_i = yield in the plot. We also know x_i = fertilizer applied to the plot. We know that $\bar{x}_{\mathcal{U}} = 100$. Here are the data:

x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$
50	1410				
100	1690				
150	1680				
200	1850				

Example: The Florida Game and Fish Commission is interested in estimating the average weight of alligators. They plan to use alligator length as an auxiliary variable, because it is much easier to observe. In fact, they've already collected so much length data they are confident that $\bar{x}_{\mathcal{U}} \approx 90$. They capture and measure $n = 22$ alligators and measure the length (inches) and weight (pounds) on each one. The data are in 'alligator.csv.' What don't we have?

We'll estimate the average alligator weight a few different ways: using summary statistics calculated in R, using the `lm` function in R to fit the regression, and using the `survey` package. In the `survey` package, there is a new function, `svyglm`. It has the general syntax:

```
svyglm(y ~ x, design = design)
```

Like with ratio estimation, we can then use the `predict` function to estimate $\bar{y}_{\mathcal{U}}$ and t_y.

Using Summary Statistics:

```
alligator<-read.csv("alligator.csv",header=TRUE)
head(alligator)
```

```
  Alligator Length Weight
1         1     94    130
2         2     74     51
3         3     82     80
4         4     58     28
5         5     86     80
6         6     94    110
```

```
n<-nrow(alligator)

##### Summary Stats ############

xbar<-mean(alligator$Length)
xbar
```

```
[1] 78.86364
```

```
ybar<-mean(alligator$Weight)
ybar
```

```
[1] 68.27273
```

```
varx<-var(alligator$Length)
varx
```

```
[1] 117.3615
```

```
vary<-var(alligator$Weight)
vary
```

```
[1] 819.8268
```

```
corrxy<-cor(alligator$Length,alligator$Weight)
corrxy
```

```
[1] 0.9180046
```

```
bhat1<-corrxy*sqrt(vary)/sqrt(varx)
bhat0<-ybar-bhat1*xbar
bhat1
```

```
[1] 2.426292
```

```
bhat0
```

```
[1] -123.0735
```

```
xbarU<-90

ybarhatreg<-ybar+bhat1*(xbarU-xbar)
ybarhatreg
```

```
[1] 95.2928
```

```
alligator$resid<-alligator$Weight - (bhat0+bhat1*alligator$Length)
s2e<-sum(alligator$resid**2)/(n-1)
s2e
```

```
[1] 128.9322
```

Using the `lm` function in R: We can fit the regression model directly in R

```
alreg<-lm(Weight~Length,data=alligator)
alreg
```

```
Call:
lm(formula = Weight ~ Length, data = alligator)

Coefficients:
(Intercept)       Length
   -123.074        2.426
```

```
anova(alreg)
```

```
Analysis of Variance Table

Response: Weight
          Df  Sum Sq Mean Sq F value    Pr(>F)
Length     1 14508.8 14508.8  107.17 1.762e-09 ***
Residuals 20  2707.6   135.4
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Using the survey package:

```
library(survey)
aldesign<-svydesign(id=~1, weights=~1, data=alligator)

alreg2<-svyglm(Weight~Length,design=aldesign)
summary(alreg2)
```

```
Call:
svyglm(formula = Weight ~ Length, design = aldesign)

Survey design:
svydesign(id = ~1, weights = ~1, data = alligator)

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -123.0735    20.6956  -5.947 8.14e-06 ***
Length         2.4263     0.2642   9.184 1.30e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 128.9322)

Number of Fisher Scoring iterations: 2
```

```
xbarU<-data.frame(Length=90)

predict(alreg2,xbarU)
```

```
    link     SE
1 95.293 4.0207
```

```
confint(predict(alreg2,xbarU),df=n-2)
```

```
     2.5 %   97.5 %
1 86.90577 103.6798
```

5.3 Domain Estimation

Sometimes we're interested in estimating characteristics (means, totals, proportions) for subpopulations. But wait, don't we already know how to do this? We estimated subpopulation characteristics when we discussed stratified sampling.

Here's what is different: now we're considering the situation where the sampling design did not include stratification. That is,

This is called **domain estimation**, and is a special case of ratio estimation.

Example: We are interested in comparing transportation challenges in low-income households with other households. A sampling frame of households is available, but we don't have information on household income.

Our first inclination might be to use the sample mean for the domain

But, n_d for each domain is not fixed by the design. n_d was unknown prior to drawing the sample. This means the denominator is a random variable.

Instead, we'll use ratio estimation (with some new variables u_i and x_i) to estimate domain means, totals, and proportions.

Let's set up the framework for domain estimation, and some new notation. We'll divide the population into D mutually exclusive and exhaustive domains.

We'll denote our domain sizes as

After collecting data, we can observe/calculate the number of sampling units in each domain.

In domain estimation:

-
-
-

We can contrast this with what we do know:

-
-

So, of course we have some new notation:

- $\mathcal{U}_d =$
- $\mathcal{S}_d =$
- $\bar{y}_{\mathcal{U}_d} =$
- $t_{\mathcal{U}_d} =$

These are population values, and $i \in \mathcal{U}_d$ means

Example: (book problem #4) A SRS of $n = 1500$ licensed boat owners was sampled from a list of $N = 400,000$ names of currently licensed boat owners. There are two domains of interest: owners of large motorboats (over 16 feet) and boat owners who own something other than a large motorboat. After collecting data, the researchers observed they had sampled $n_1 = 472$ owners in domain 1 and $n_2 = 1028$ owners in domain 2. The boat owners were asked to report $y_i =$ how many children they have. We want to estimate:

- the mean number of children for boat owners in domain 1
- the total number of children for large boat owners

Now, the population domain mean is

We're going to need some new variables. The first one really isn't new. We're going to construct our denominator using

We can still think of this as an auxiliary variable!

So, for domain 1 in our example:

For our numerator, we'll define

In our example,

So, our population ratio is now

and our ratio estimator for $\bar{y}_{\mathcal{U}_d}$ is

Example: So, to summarize y_i, u_i, and x_i in our example:

So, an excerpt of the full data set might look something like this:

Owner	Domain	Num. Kids (y_i)	Numerator (u_i)	Denominator (x_i)
1	1	3		
2	1	2		
3	2	5		
4	1	0		
5	2	0		
6	2	1		
7	1	1		
8	2	2		
⋮				

If we focus on Domain 1 (large motorboats), we can summarize the data as follows:

y_i	Num. Respondents
0	1104
1	139
2	166
3	63
4	19
5	5
6	3
8	1
Total	1500

So,

$$\bar{u} =$$

and

$$\bar{x} =$$

Then,

We do still need to figure out the estimated variance!

When we initially talked about ratio estimation, we noted that

So now,

If n_d and n are large then $n_d - 1 \approx n_d$ and $n \approx n - 1$, so

Example:

Suppose instead we are interested in estimating the total for a domain:

There are two possibilities:

1. N_d is known

2. N_d is unknown

Example: We don't know the total number of boat owners that fall into Domain 1.

We can construct confidence intervals for domain estimates

We can also compare domains using hypothesis testing!

We can use R to carry out domain estimation. It's easiest to use the `subset` or `svyby` functions in the `survey` package to do so. If we've already set up our design using `svydesign` we can use the `subset` function to redefine the design for the domain of interest. The generic syntax is

```
newdesign<-subset(design,domain)
```

Let's check our boats estimates in R. The first thing we need to do is get the data ready for analysis.

```
boats <- data.frame(
  numkids=c(0,1,2,3,4,5,6,8),
  dom1=c(76,139,166,63,19,5,3,1),
  dom2=c(528,189,225,76,8,2,0,0))

boats
```

```
  numkids dom1 dom2
1       0   76  528
2       1  139  189
3       2  166  225
4       3   63   76
5       4   19    8
6       5    5    2
7       6    3    0
8       8    1    0
```

```
boatdom1<-boats[rep(1:nrow(boats),times=boats$dom1),]
boatdom1$dom<-1
boatdom2<-boats[rep(1:nrow(boats),times=boats$dom2),]
boatdom2$dom<-2
boatdf<-rbind(boatdom1,boatdom2)
dim(boatdf)
```

```
[1] 1500    4
```

```
head(boatdf)
```

```
    numkids dom1 dom2 dom
1         0   76  528   1
1.1       0   76  528   1
1.2       0   76  528   1
1.3       0   76  528   1
1.4       0   76  528   1
1.5       0   76  528   1
```

```
N<-400000
n<-nrow(boatdf)
boatdf$sampwt<-N/n
head(boatdf)
```

```
    numkids dom1 dom2 dom   sampwt
1         0   76  528   1 266.6667
1.1       0   76  528   1 266.6667
1.2       0   76  528   1 266.6667
1.3       0   76  528   1 266.6667
1.4       0   76  528   1 266.6667
1.5       0   76  528   1 266.6667
```

Now we can find the estimates:

```
library(survey)

boatdesign<-svydesign(id=~1,weights=~sampwt,fpc=rep(N,n),data=boatdf)
dlarge<-subset(boatdesign,dom=='1')
smean1<-svymean(~numkids,design=dlarge)
smean1
```

```
          mean     SE
numkids 1.6674 0.0543
```

```
stot1<-svytotal(~numkids,dlarge)
stot1
```

```
         total    SE
numkids 209867 10510
```

```
bothmean<-svyby(~numkids,by=~factor(dom),design=boatdesign,svymean)
bothmean
```

```
  factor(dom)   numkids         se
1           1 1.6673729 0.05429457
2           2 0.8842412 0.03289729
```

5.4 Poststratification

We mentioned **poststratification** earlier as a reason we might want to use ratio estimation, as it allows us to adjust for demographics or similar. How is it different from stratification?

In the study process, poststratification fits in like this:

Before the study: Information is either not available on poststrata membership, or it's not used in designing the study. Information is available on poststrata sizes (N_h) and these values are fixed.

During the study: Obtain information on poststrata membership, along with other responses.

After the study: Data includes poststrata membership and responses for each sampling unit. Researcher has n_h, which are random. Can now estimate poststratum sample means (totals) and pool them for an overall population estimate based on poststratification weights.

Here's the set up

The estimation process is basically the same as domain estimation, but now $N_1, \ldots, N_h$ are known and the values can be used in estimation. So, for example,

Define

In ratio estimation, weights were calibrated to add to t_x (we saw this with the small fish/nets examples). Now,

So we can get ratio estimates for the poststrata totals (means)

and then pool across them just like we did in stratified sampling

Remember that stratification was not built into the sampling design, so if an SRS was used as the design and n_h (and thus n) is pretty large, we can use the variance from proportional allocation:

Often, you'll hear pollsters talk about weighting the data. What they are doing is using poststratification weights. As mentioned above, poststratification modifies the sampling weights so they sum to N_h in poststratum h. Remember that in an SRS,

and in stratified

Now, based on the way we've defined $x_{ih} =$

So we can write our poststratification population estimates the same way we did with simple random samples back in Chapter 3:

Example: Let's revisit the faculty publication data we used back in Chapter 4, and suppose that the data were collected as a SRS. The researchers knew how many faculty fell into each discipline class, but only had faculty names and emails available for the sampling frame. They drew a random sample of $n = 50$ faculty members from the $N = 807$ and asked them about both their discipline and the number of publications they had in the last year.

Stratum	N_h	n_h	$\widehat{N}_h$	w_i	$w_i^* = \frac{N_h}{\widehat{N}_h} w_i$	$\bar{y}_h$	s_h^2
Bio Sci	102	7				3.14	6.81
Phy Sci	310	19				2.105	8.21
Soc Sci	217	13				1.231	4.359
Humanities	178	11				0.4545	0.8727
Total	807	50					

I set up the data as we did in Chapter 4. We can check our work use the `postStratify` function in the `survey` package. We set up the design in `svydesign` assuming a SRS:

```
N<-807
n<-nrow(pubsdf)
pubsdesign<-svydesign(id=~1,weights=rep(N/n,n),fpc=rep(N,n),data=pubsdf)
```

and then tell R the size of each poststratum:

```
pop.disc<-data.frame(discipline=c("Biological","Physical","Social","Humanities"),
                     Freq=c(102,310,217,178))
pop.disc
```

```
  discipline Freq
1 Biological  102
2   Physical  310
3     Social  217
4 Humanities  178
```

We can use `postStratify` to set up the poststratification weights.

```
pubsdesps<-postStratify(pubsdesign,~discipline,pop.disc)
summary(pubsdesps)
```

```
Independent Sampling design
postStratify(pubsdesign, ~discipline, pop.disc)
Probabilities:
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.05991 0.06025 0.06129 0.06207 0.06180 0.06863
Population size (PSUs): 807
Data variables:
[1] "discipline" "pubs"       "Nh"         "samp"
```

```
1/unique(pubsdesps$prob)
```

```
[1] 14.57143 16.31579 16.69231 16.18182
```

Then once those weights are set up, we can use `svymean` and `svytotal` just like before

```
svymean(~pubs,pubsdesps)
```

```
       mean     SE
pubs 1.6372 0.3088
```

```
svytotal(~pubs,pubsdesps)
```

```
      total    SE
pubs 1321.2 249.2
```

We can compare these to the SRS estimates

```
svymean(~pubs,pubsdesign)
```

```
     mean     SE
pubs 1.66 0.3316
```

```
svytotal(~pubs,pubsdesign)
```

```
      total     SE
pubs 1339.6 267.61
```

5.5 Ratio Estimation with Stratified Sampling

Everything we've done so far with ratio and regression estimations assumed the sampling design was SRS. But, these concepts can be applied to any sampling design, including stratified sampling. We just have to set up the design to include the stratification.

Example: On Homework 4, you used the auxiliary variable of x_i = number of acres devoted to farming in 1987 along with the response y_i = number of acres devoted to farming in 1992. There's another version of this data set (`agstrat`) on Dr. Lohr's website that includes a stratification variable: region (North Central, Northeast, South, and West). The manual on using R to carry out these analyses linked in Canvas shows example code implementing ratio estimation with stratification using this data set.

6 Cluster Sampling

So far we've talked about two sampling designs in detail: simple random sampling (SRS) and stratified random sampling. Sometimes, however, it's impossible (or at least very difficult) to collect an SRS or stratified random sample. This could be due to many things, including

- cost (in time or money)
- lack of a sampling frame for the target population from which to draw our sample

We'll consider a couple of examples to demonstrate this.

Example: About a decade ago, the journal *Basic and Applied Social Psychology* banned p-values from its publication, and this sparked a lot of thought, articles, and talks about the role of p-values in science. Suppose we want to estimate the proportion of articles in the social and behavioral sciences literature that contained p-values in 2023. There are 1285 such journals. There are a couple of options:

-

-

Example: Suppose we want to estimate the proportion of Nebraska high school students who are taking (or have taken) a dual credit course.

Both of these are examples of **cluster samples**. In cluster sampling,

Population of Interest	Elements secondary units	Cluster primary units
Articles in social/behavioral journals		
US residents		
Lincoln households		
UNL staff members		
Maple trees in Vermont		

So, a **cluster sample** is a probability sample in which a sampling unit is a cluster. Sometimes the cluster is determined by the availability of a sampling frame.

Frame	Sampling Unit	Element (Observation Unit)

This might at first blush look like stratification. Both clustering and stratification group population members. But, selection is totally different. Additionally,

Example: High school dual credit

6.1 One-Stage Cluster Sampling

We're going to start with **one-stage cluster sampling**. One-stage means

Here's the set-up:

- Divide the population into N clusters, also known as **primary sampling units** (PSUs).
- Cluster i has size M_i, so M_i is the number of **secondary sampling units** (SSUs).

- Clusters can be selected using any sampling design. We'll assume an SRS.
- On each SSU in a selected PSU, the characteristic of interest is observed. We are back to 2 subscripts:

We've got lots of new notation, which we'll break up by sampling level in addition to population/sample.

Population Quantities: PSU level

N	t_i
M_i	t
M_0	σ_t^2

Population Quantities: SSU level

$\bar{y}_{\mathcal{U}} = \frac{1}{M_0} \sum_{i=1}^{N} \sum_{j=1}^{M_i} y_{ij}$

$\bar{y}_{i\mathcal{U}} = \frac{1}{M_i} \sum_{j=1}^{M_i} y_{ij} = \frac{t_i}{M_i}$

$\sigma^2 = \frac{1}{M_0 - 1} \sum_{i=1}^{N} \sum_{j=1}^{M_i} (y_{ij} - \bar{y}_{\mathcal{U}})^2$

$\sigma_i^2 = \frac{1}{M_i - 1} \sum_{j=1}^{M_i} (y_{ij} - \bar{y}_{i\mathcal{U}})^2$

Sample Quantities

$\mathcal{S}$ = sample of PSUs from the whole population

$\mathcal{S}_i$ = sample of SSUs selected from PSU i

n = number of PSUs in the sample

m_i = number of SSUs from PSU i in the sample

$\bar{y}_i = \frac{1}{m_i} \sum_{j \in \mathcal{S}_i} y_{ij}$

$\hat{t}_i = \frac{M_i}{m_i} \sum_{j \in \mathcal{S}_i} y_{ij}$

$s_t^2 = \frac{1}{n-1} \sum_{i \in \mathcal{S}} \left(\hat{t}_i - \frac{\sum_{i \in \mathcal{S}} \hat{t}_i}{n} \right)^2$

$s_i^2 = \frac{1}{m_i - 1} \sum_{j \in \mathcal{S}_i} (y_{ij} - \bar{y}_i)^2$

w_{ij} = sampling weight for SSU j in PSU i

What does cluster sampling look like?

Now, because we are considering 1-stage cluster sampling

6.1.1 Equal size clusters

We're going to start with the simplest case: clusters of equal size. This means

If clusters are equal size, estimating population means and totals is easy–we treat the PSU mean/total as the observation and ignore the individual elements.

The clusters are a SRS of the population, so we can use the variances from the SRS

Dealing with means is also straightforward

Example: Inspecting cases of canned goods for damaged cans. We have a truckload of 100 cases, each with 12 cans. We select a SRS of 4 cases. Here's the data.

Can #	Case 1	Case 2	Case 3	Case 4
1	0	0	0	0
2	0	0	0	0
3	1	0	0	0
4	0	0	1	0
5	0	0	0	0
6	1	1	0	0
7	0	0	0	1
8	0	0	0	0
9	0	0	1	0
10	0	0	0	0
11	0	1	0	0
12	0	0	0	0
t_i				

Suppose while checking the cans for damage the cans were also weighed, to make sure they are labeled correctly. The cans are labeled 8 oz.

Can #	Case 1	Case 2	Case 3	Case 4
1	8.15	8.03	7.65	7.99
2	7.70	8.16	7.90	7.79
3	7.76	7.73	7.84	7.96
4	7.59	7.67	8.14	8.20
5	7.71	7.92	8.07	7.92
6	7.55	7.94	7.83	7.98
7	8.30	7.93	8.07	7.67
8	8.26	7.77	7.96	7.99
9	8.01	7.93	7.96	7.85
10	8.06	8.01	7.91	8.13
11	7.69	7.88	7.95	7.56
12	8.08	7.58	7.79	7.94
t_i				

Like with previous designs, each unit has a **sampling weight**. Sampling weights are associated with SSUs.

Let w_{ij} be the weight for observation unit (SSU) j in PSU i.

$$\sum_{i\in\mathcal{S}}\sum_{j=1}^{M} w_{ij} = \sum_i\sum_j \frac{N}{n} = \sum_i \frac{N}{n}\sum_j 1 = \sum_i \frac{N}{n}(M) = \frac{NM}{n}\sum_{i\in\mathcal{S}} 1 = \frac{NM}{n}(n) = NM$$

Sample weights can be used just like they are in the SRS to calculate estimators:

$$\hat{t} = \sum_i\sum_j w_{ij}y_{ij} \text{ and } \hat{\bar{y}} = \frac{\sum_i\sum_j w_{ij}y_{ij}}{\sum_i\sum_j w_{ij}}$$

Survey packages use these weights to get the estimates, just as with other designs.

In R, the main thing that will change is in the `svydesign` statement. The `id` option identifies clusters, so now we will use that. With previous designs, our R code looked like:

```
design<-svydesign(id=~1,weights=~sampwt,fpc=rep(N,n),data=dataset)
```

with potentially a `strata=` option included. Now, our generic syntax will be

```
design<-svydesign(id=~cluster,weights=~sampwt,fpc=~fpc,data=dataset)
```

where `cluster` is the column identifying the PSU in the data set.

Example: For the cans data set, I used the code

```
cans<-read.csv("cans.csv",header=TRUE)

N<-100
n<-4
cans$sampwt<-N/n
head(cans)
```

```
  weight case sampwt
1   8.15    1     25
2   7.70    1     25
3   7.76    1     25
4   7.59    1     25
5   7.71    1     25
6   7.55    1     25
```

```
library(survey)
candesign<-svydesign(id=~case,weights=~sampwt,fpc=~rep(100,48),data=cans)
candesign
```

```
1 - level Cluster Sampling design
With (4) clusters.
svydesign(id = ~case, weights = ~sampwt, fpc = ~rep(100, 48),
    data = cans)
```

```
wtmean<-svymean(~weight,candesign)
wtmean
```

```
         mean     SE
weight 7.9054 0.0093
```

6.1.2 Unequal size clusters

Often clusters are not the same size (for example, household sizes) so frequently we need to account for the cluster size. Recall that

We could use the same (unbiased) estimators we did when cluster sizes are all equal:

$$\hat{t}_{unb} = \frac{N}{n}\sum_{i\in\mathcal{S}} t_i \text{ with } \text{SE}(\hat{t}_{unb}) = N\sqrt{\left(1-\frac{n}{N}\right)\frac{s_t^2}{n}}$$

and

$$\hat{\bar{y}} = \frac{\hat{t}_{unb}}{M_0} \text{ with } \text{SE}(\hat{\bar{y}}) = \frac{N}{M_0}\sqrt{\left(1-\frac{n}{N}\right)\frac{s_t^2}{n}}$$

These aren't usually the estimators we use, though. Why not? Why is it important to account for cluster sizes? Typically, cluster total t_i is correlated with cluster size M_i

We know how to deal with this situation!

So now, M_i serves as our auxiliary variable and our response is t_i. We can then use our ratio estimate for population mean, and its corresponding variance estimate

$$\hat{\bar{y}}_r = \frac{\sum_{i=1}^n t_i}{\sum_{i=1}^n M_i}$$

with

$$\begin{aligned}
\widehat{\text{Var}(\hat{\bar{y}}_r)} &= \left(1 - \frac{n}{N}\right) \frac{1}{n\bar{M}_{\mathcal{U}}^2} \frac{\sum_{i=1}^n (t_i - \hat{\bar{y}}_r M_i)^2}{n-1} \\
&= \left(1 - \frac{n}{N}\right) \frac{1}{n\bar{M}_{\mathcal{U}}^2} \frac{\sum_{i=1}^n M_i^2 (\bar{y}_i - \hat{\bar{y}}_r)^2}{n-1}
\end{aligned}$$

Example: Can weights, revisited.

Can #	Case 1	Case 2	Case 3	Case 4
1	8.15	8.03	7.65	7.99
⋮	⋮	⋮	⋮	⋮
12	8.08	7.58	7.79	7.94
t_i	94.86	94.55	95.07	94.98

We can then easily get an estimate for the population total as well as an estimate of variance.

Example:

Example: (adapted from Example 5.6 in the book) A school district in a large city has $N = 187$ algebra classes. A researcher takes an SRS of $n = 4$ classes and each student is given a quiz. In our sample, class sizes range from 17 to 34.

Class	M_i	$\bar{y}_i$	t_i	$M_i^2(\bar{y}_i - \hat{\bar{y}}_r)^2$
1	34	58		
2	28	64.9	1816	899.28
3	32	72.1	2308	70051.268
4	17	58.2	989	9157.150
	111		7085	119385.389

We can extend the idea of sample weights to different cluster sizes. We're still in 1-stage clustering, so all SSUs are observed for a selected PSU. This means

$$w_{ij} = \frac{1}{\text{P(SSU } j \text{ in PSU } i \text{ is in the sample)}} = \frac{1}{n/N} = \frac{N}{n}$$

When all clusters are the same size

$$\sum_{i \in \mathcal{S}} \sum_{j \in \mathcal{S}_i} w_{ij} = NM = M_0$$

But when the PSUs have different sizes, $\sum_i \sum_j w_{ij}$ estimates population size:

$$\sum_{i \in \mathcal{S}} \sum_{j \in \mathcal{S}_i} w_{ij} = \sum_i \sum_j \frac{N}{n} = \sum_{i \in \mathcal{S}} \frac{N}{n} \sum_{j \in \mathcal{S}_i} 1 = \sum_{i \in \mathcal{S}} \frac{N}{n} M_i = \frac{N}{n} \sum_{i \in \mathcal{S}} M_i = \hat{M}_0$$

The weights can be used to get estimators in exactly the same way, and this is what survey packages do.

6.2 Two-Stage Cluster Sampling

In **two-stage cluster sampling**, not all of the SSUs are observed for a selected PSU. Instead,

- Stage 1: Select clusters (PSUs)
- Stage 2: Select elements within each selected cluster (SSUs)

and we only collect data on the SSUs that were sampled from the cluster. Here's what two-stage cluster sampling looks like

Now, because there are two stages of sampling, there are two sets of sampling units.

- Stage 1: Select SRS of n PSU from the population of N PSUs
- Stage 2: Select SRS of m_i SSUs from the M_i elements of PSU i

What are the implications?

With one-stage cluster sampling, we had data on every single element in the selected clusters

$$\Rightarrow \text{Cluster population values are known. We directly observe } \bar{y}_i, t_i$$

But with two-stage cluster sampling, we only observe data on a sample of the elements in a selected cluster.

$$\Rightarrow \text{We have to calculate estimates of } \bar{y}_i, t_i$$

So why use two-stage cluster sampling?

- Maybe we only have access to a frame that lists clusters
- Elements in a PSU are so similar that it's a waste of resources to sample all of them. Extra effort does not provide much additional information.

As you can maybe guess, the variance estimates in two-stage cluster sampling are a LOT messier because we have to account for variance among the PSU (like before) but also the variance among the SSUs within the PSUs.

Let's consider the unbiased estimator of the population total

$$\hat{t}_{unb} = \frac{N}{n} \sum_{i \in \mathcal{S}} t_i$$

We now no longer know t_i, and we have to estimate it. We've observed data on m_i elements in cluster i, so

$$\hat{t}_i = M_i \bar{y}_i = \sum_{j=1}^{m_i} \frac{M_i}{m_i} y_{ij}$$

Basically, we're using the same estimator we did with one-stage clustering, but plugging in estimated cluster totals rather than known cluster totals.

- One-stage: $\hat{t}_{unb} = \frac{N}{n} \sum_i \underbrace{t_i}_{\text{known}} = \frac{N}{n} \sum_i M_i \underbrace{\bar{y}_{iU}}_{\text{known}}$
- Two-stage: $\hat{t}_{unb} = \frac{N}{n} \sum_i \underbrace{\hat{t}_i}_{\text{est}} = \frac{N}{n} \sum_i M_i \underbrace{\bar{y}_i}_{\text{est from } m_i \text{ elements}}$

The variance of $\hat{t}_{unb}$ has two components, associated with the 2 sampling stages: variation among PSUs and variation among SSUs within PSUs:

- One-stage: only variation among PSUs

$$\widehat{\text{Var}(\hat{t})} = N^2 \left(1 - \frac{n}{N}\right) \frac{s_t^2}{n}, \quad s_t^2 = \frac{1}{n-1} \sum_i \left(\underbrace{t_i}_{\text{known}} - \frac{\hat{t}}{N} \right)^2$$

- Two-stage: we now have to estimate t_i with $\hat{t}_i$

$$\widehat{\text{Var}(\hat{t}_{unb})} = \underbrace{N^2 \left(1 - \frac{n}{N}\right) \frac{s_t^2}{n}}_{\text{among PSUs}} + \underbrace{\frac{N}{n} \sum_{i \in \mathcal{S}} \left(1 - \frac{m_i}{M_i}\right) M_i^2 \frac{s_i^2}{m_i}}_{\text{within PSUs}}$$

where $s_t^2 = \frac{1}{n-1} \sum_{i \in \mathcal{S}} \left(\hat{t}_i - \frac{\hat{t}_{unb}}{N} \right)^2$ and $s_i^2 = \frac{1}{m_i - 1} \sum_{j \in \mathcal{S}_i} (y_{ij} - \bar{y}_i)^2$.

Example: Let's go back to the can example, and suppose we weighed only 3 cans in each case.

Can #	Case 1	Case 2	Case 3	Case 4
1	8.15	8.03	7.65	7.99
2	7.70	8.16	7.90	7.79
3	8.08	7.58	7.79	7.94
	23.93	23.77	23.34	23.72

- Stage 1:
  - Cluster = Case
  - $N = 100$
  - $n = 4$
- Stage 2:
  - Element = Can
  - $M_i = 12$
  - $m_i = 3$

Can #	Case 1	Case 2	Case 3	Case 4
1	8.15	8.03	7.65	7.99
2	7.70	8.16	7.90	7.79
3	8.08	7.58	7.79	7.94
$\bar{y}_i$	7.977	7.923	7.78	7.917
$\hat{t}_i = M_i \bar{y}_i$	95.724	95.076	93.36	95.004
$s_i^2 = \frac{1}{m_i - 1} \sum_{j=1}^{m_i} (y_{ij} - \bar{y}_i)^2$	0.0586	0.0926	0.0157	0.01099

So,

$$\hat{t}_{unb} = \frac{N}{n} \sum_{i=1}^{n} \hat{t}_i = \frac{100}{4}[95.724 + \cdots + 95.004] = 9479.85$$

and

$$s_t^2 = \frac{1}{n-1} \sum_{i \in \mathcal{S}} \left(\hat{t}_i - \frac{\hat{t}_{unb}}{N} \right)^2 = \frac{1}{3} \left[\left(95.724 - \frac{9479.85}{100} \right)^2 + \cdots + \left(95.004 - \frac{9479.85}{100} \right)^2 \right] = 1.015$$

So,

$$\begin{aligned}
\widehat{\text{Var}(\hat{t}_{unb})} &= \underbrace{N^2 \left(1 - \frac{n}{N} \right) \frac{s_t^2}{n}}_{\text{among PSUs}} + \underbrace{\frac{N}{n} \sum_{i \in \mathcal{S}} \left(1 - \frac{m_i}{M_i} \right) M_i^2 \frac{s_i^2}{m_i}}_{\text{within PSUs}} \\
&= (100)^2 \left(1 - \frac{4}{100} \right) \frac{1.015}{4} + \frac{100}{4} \left[\left(1 - \frac{3}{12} \right) (12)^2 \frac{0.0586}{3} + \cdots + \left(1 - \frac{3}{12} \right) (12)^2 \frac{0.01099}{3} \right] \\
&= 2436.055 + 480.303 = 2916.358 \Rightarrow \text{SE} = 54.0033
\end{aligned}$$

We can use this estimator for the total (and its estimated variance) to get an estimator for the population mean $\bar{y}_{\mathcal{U}}$.

$$\hat{\bar{y}}_{unb} = \frac{\hat{t}_{unb}}{M_0}$$

$$\widehat{\text{Var}(\bar{y}_{unb})} = \frac{1}{M_0^2}\widehat{\text{Var}(\hat{t}_{unb})}$$

Example: For the can data,

$$\hat{\bar{y}}_{unb} = \frac{\hat{t}_{unb}}{M_0} = \frac{9479.85}{1200} = 7.8999$$

and

$$\widehat{\text{Var}(\bar{y}_{unb})} = \frac{1}{M_0^2}\widehat{\text{Var}(\hat{t}_{unb})} = \frac{1}{1200^2}2596.156 = 0.0018$$

We can also use ratio estimation like we did with one-stage cluster sampling. The ratio estimate for the population mean $\bar{y}_{\mathcal{U}}$ is

$$\hat{\bar{y}}_r = \frac{\sum_{i=1}^n \hat{t}_i}{\sum_{i=1}^n M_i} = \frac{\sum_i M_i \bar{y}_i}{\sum_i M_i}$$

with estimated variance

$$\widehat{\text{Var}(\hat{\bar{y}}_r)} = \frac{1}{\bar{M}_{\mathcal{U}}^2}\left[\left(1-\frac{n}{N}\right)\frac{s_r^2}{n} + \frac{1}{nN}\sum_{i=1}^n M_i^2\left(1-\frac{m_i}{M_i}\right)\frac{s_i^2}{m_i}\right]$$

where

$$s_r^2 = \frac{1}{n-1}\sum_{i=1}^n (M_i\bar{y}_i - M_i\hat{\bar{y}}_r)^2 = \frac{1}{n-1}\sum_{i=1}^n M_i^2(\bar{y}_i - \hat{\bar{y}}_r)^2.$$

Example:

Can #	Case 1	Case 2	Case 3	Case 4
1	8.15	8.03	7.65	7.99
2	7.70	8.16	7.90	7.79
3	8.08	7.58	7.79	7.94
$\bar{y}_i$	7.977	7.923	7.78	7.917
$\hat{t}_i = M_i\bar{y}_i$	95.724	95.076	93.36	95.004
$s_i^2 = \frac{1}{m_i-1}\sum_{j=1}^{m_i}(y_{ij}-\bar{y}_i)^2$	0.0586	0.0926	0.0157	0.01099

So,

$$\hat{\bar{y}}_r = \frac{\sum_{i=1}^n \hat{t}_i}{\sum_{i=1}^n M_i} = \frac{95.724 + \cdots + 95.004}{12 + \cdots + 12} = 7.899$$

and

$$s_r^2 = \frac{1}{n-1}\sum_{i=1}^n M_i^2(\bar{y}_i - \hat{\bar{y}}_r)^2 = \frac{1}{3}\left[12^2(7.977 - 7.899)^2 + \cdots + 12^2(7.917 - 7.899)^2\right] = 1.01496$$

and therefore

$$\begin{aligned}
\widehat{\mathrm{Var}(\hat{\bar{y}}_r)} &= \frac{1}{\bar{M}_U^2}\left[\left(1 - \frac{n}{N}\right)\frac{s_r^2}{n} + \frac{1}{nN}\sum_{i=1}^n M_i^2\left(1 - \frac{m_i}{M_i}\right)\frac{s_i^2}{m_i}\right] \\
&= \frac{1}{12^2}\left[\left(1 - \frac{4}{100}\right)\frac{1.01496}{4} + \frac{1}{(4)(100)}(19.2121)\right] = 0.00205
\end{aligned}$$

so the SE is 0.045.

We can use the ratio estimate of the population mean to get the ratio estimate for the population total

$$\hat{t}_r = M_0\hat{\bar{y}}_r \text{ with } \widehat{\mathrm{Var}(\hat{t}_r)} = M_0^2\widehat{\mathrm{Var}(\hat{\bar{y}}_r)}$$

In the can example,

$$\hat{t}_r = M_0\hat{\bar{y}}_r = (1200)(7.899) = 9478.8$$

and

$$\widehat{\mathrm{Var}(\hat{t}_r)} = M_0^2\widehat{\mathrm{Var}(\hat{\bar{y}}_r)} = (1200^2)(0.002025) = 2916.207$$

Why do the unbiased estimators and their variances match so closely with the (biased) ratio estimators? There is little variance among cases and cans and the clusters are all the same sizes.

When should we use the unbiased estimators versus the (biased) ratio estimators? Some things to keep in mind:

- Unbiased estimation requires that we know N and M_0.
- Ratio estimation only requires knowledge of M_i
- Unbiased estimation can have poor precision (big variance) if
  - Cluster sizes (M_i) are unequal
  - Cluster totals (t_i) are roughly proportional to cluster size

- Ratio (biased) estimation can be precise if
  - t_i is roughly proportional to M_i

Example: (Example 5.9 in book) We want to estimate the average number of legs on healthy puppies in animal shelters. There are two animal shelters in the town. Shelter A has 30 puppies and Shelter B has 10 puppies. We select one shelter at random. After selecting the shelter, we'll select 2 puppies at random.

Pick Shelter A: $y_{A1} = 4$, $y_{A2} = 4$

$$\begin{aligned} \hat{t}_A &= M_A \bar{y}_A = (30)(4) = 120 \\ \hat{t}_{unb} &= \frac{N}{n}\sum_i \hat{t}_i = \frac{2}{1}(120) = 240 \\ \Rightarrow \hat{\bar{y}}_{unb} &= \frac{\hat{t}_{unb}}{M_0} = \frac{240}{40} = 6 \text{ puppy legs!} \end{aligned}$$

Pick Shelter B: $y_{B1} = 4$, $y_{B2} = 4$

$$\begin{aligned} \hat{t}_B &= M_B \bar{y}_B = (10)(4) = 40 \\ \hat{t}_{unb} &= \frac{N}{n}\sum_i \hat{t}_i = \frac{2}{1}(40) = 80 \\ \Rightarrow \hat{\bar{y}}_{unb} &= \frac{\hat{t}_{unb}}{M_0} = \frac{80}{40} = 2 \text{ puppy legs!} \end{aligned}$$

So the estimator is unbiased (6+2/2=4), but gives ridiculous estimates (not shown, but the variance of the unbiased mean here is 6400).

The ratio estimator on the other hand,

Shelter A: $\hat{\bar{y}}_r = \frac{\sum_i \hat{t}_i}{\sum_i M_i} = \frac{120}{30} = 4$

Shelter B: $\hat{\bar{y}}_r = \frac{\sum_i \hat{t}_i}{\sum_i M_i} = \frac{40}{10} = 4$, and the variance is 0.

Let's go back to the estimator for the population total

$$\begin{aligned}\hat{t}_{unb} &= \frac{N}{n}\sum_i \hat{t}_i = \frac{N}{n}\sum_i M_i\bar{y}_i = \frac{N}{n}\sum_i \frac{M_i}{m_i}\sum_{j=1}^{m_i} y_{ij} \\ &= \sum_i\sum_j \frac{N}{n}\frac{M_i}{m_i}y_{ij} = \sum_i\sum_j w_{ij}y_{ij}\end{aligned}$$

So, the weight for element j in cluster i is

$$w_{ij} = \frac{N}{n}\frac{M_i}{m_i}$$

Is this still an equal probability sample? Let's look more closely as the inclusion probabilities

- $\pi_i = \text{P(cluster } i \text{ is in the sample)} = \frac{n}{N}$ because clusters are selected in a SRS
- $\pi_{j|i} = \text{P(element } j \text{ is selected GIVEN cluster } i \text{ is in the sample)} = \frac{m_i}{M_i} = \frac{\text{total selected}}{\text{total in cluster}}$
- $\pi_{ij} = \text{P(element } j \text{ and cluster } i \text{ is in the sample)} = \pi_i\pi_{j|i} = \left(\frac{n}{N}\right)\left(\frac{m_i}{M_i}\right) = \frac{nm_i}{NM_i} \Rightarrow w_{ij} = \frac{NM_i}{nm_i}$

This is only constant if M_i/m_i is constant, $M_i/m_i = c$.

So, if we want a self-weighting design with two-stage clustering

- Stage 1: Select n PSUs from N population PSUs using SRS
- Stage 2: Choose m_i proportional to M_i so that m_i/M_i is constant. Select the m_i SSUs using a SRS.
- The sample weight for SSU j in Cluster i is then constant for all elements

What are the implications of m_i/M_i constant? You have to subsample many more elements in big clusters than in small clusters.

We can use R to get estimates from one-stage clustering with unequal cluster sizes, as well as two-stage clustering. The differences are going to come in the svydesign statement, and the inputs we pass to it.

For one-stage clustering with unequal cluster sizes, nothing really changes from equal cluster sizes.

Example: Algebra classes. Let's now use all $n = 12$ classes that were randomly selected out of the $N = 187$, rather than just 4.

```
algebra<-read.csv("algebra.csv",header=TRUE)
head(algebra)
```

```
  class Mi score
1    23 20    57
2    23 20    90
3    23 20    56
4    23 20    57
5    23 20    46
6    23 20    55
```

```
N<-187
n<-12
m<-299
algebra$sampwt<-N/n
head(algebra)
```

```
  class Mi score   sampwt
1    23 20    57 15.58333
2    23 20    90 15.58333
3    23 20    56 15.58333
4    23 20    57 15.58333
5    23 20    46 15.58333
6    23 20    55 15.58333
```

```
algdesign<-svydesign(id=~class,weights=~sampwt,fpc=~rep(N,m),data=algebra)
algdesign
```

```
1 - level Cluster Sampling design
With (12) clusters.
svydesign(id = ~class, weights = ~sampwt, fpc = ~rep(N, m), data = algebra)
```

```
scoremean<-svymean(~score,algdesign)
scoremean
```

```
        mean     SE
score 62.569 1.4916
```

Note the fpc option uses $m = 299$, since there were a total of 299 students in our sample of 12 classes. This gives the output

For two-stage clustering we do need to adjust the code. We'll need to identify both the PSUs and the SSUs

Example: (Book Problem #8) A person with a large library needs to estimate the purchase cost and replace value of a book collection for their insurance. They have 44 shelves with books and randomly select 12 shelves. They then count the number of books M_i on each of the selected shelves, and generate five random numbers between 1 and M_i for each selected shelf to determine which books to examine. They select the specific books determined by the random numbers and look up the replacement value. The data are in `books.csv` on Canvas. We'll estimate total replacement cost for the library as well as average replacement cost per book.

In R, we'll use the code

```
    books<-read.csv('books.csv',header=TRUE)
    head(books)
```

```
  shelf Mi booknumber purchase replace
1     2 26          3        1      13
2     2 26          5        1      13
3     2 26          6        1       3
4     2 26         18        8       8
5     2 26         19        7      11
6     4 52          2        1       6
```

```
    N<-44
    n<-12
    nbooks<-nrow(books)

    booksdesign<-svydesign(id=~shelf+booknumber,fpc=~rep(N,nbooks)+Mi,nest=TRUE,data=books)
    svytotal(~replace,booksdesign)
```

```
        total     SE
replace 32638 5733.5
```

```
    svymean(~replace,booksdesign)
```

```
          mean     SE
replace 23.611 5.4759
```